

When Guarantees Go Silent: Feasibility and Attackability of Online Error Control for ML Intrusion Detection

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Abstract—Machine-learning intrusion detectors emit millions of predictions per hour, and a growing line of work makes the resulting alerts interpretable by wrapping them in a statistical trust layer: conformal evidence, aggregation into security-meaningful units, and an online false-discovery-rate controller. The constituent procedures are individually valid under their assumptions, but composing them raises separate questions: whether a useful discovery is statistically feasible at security-scale horizons, and whether the composition resists an adversary who knows it is there. Validity is not feasibility: under finite-resolution conformal evidence, an uninterrupted controller and a spending sequence that decays, the set of times at which any rejection is possible during a rejection-free run is finite and the infeasible state absorbing — for two families covering LOND, LORD++, e-LOND and e-LORD. Feasibility over a horizon T then needs $|\mathcal{C}| \geq k/\alpha_T - 1$, which under the max-min-optimal horizon-uniform sequence is $kT/c_0 - 1$ in a cold-start coefficient c_0 : 6.5×10^8 calibration flows for level- w_0 procedures (LORD++) and 3.3×10^8 for LOND/e-LOND at this dataset’s scale, and this is the cheapest sequence — horizon-free $\gamma \propto j^{-1.6}$ asks 3.2×10^{13} . Aggregation is the direct route to feasibility, at a price: coarsening the alerting unit buys feasibility on that exchange rate and pays in resolution, each issued alert blurring together more of the attack — the bridge between our two results rather than a third. The aggregation and adaptive state are then attacker-facing. No symmetric e-emerging family that can fire resists adversarial padding — a pre-committed cap escapes only by leaving the class, and forfeits measured power for it: suppressing an alert is a low-footprint evasion costing a median of 6–5,202 ordinary flows per window across the two orders we report, and 23–33 for the episodes the canonical order detects at the primary window (all lower bounds). And the escape that conditions its spending index on observed evidence exposes that index in closed form, letting an attacker silence ADDIS permanently — shown at a window whose evidence is not a valid e-value and then reproduced on a synthetic stream where ADDIS’s guarantee does hold, and conspicuous in volume rather than cheap. Evaluated on 16,353,511 live-fire flows over five deployment windows, with a second detector and a second dataset used to test whether the attack transfers, statistical trust layers for security ML must be assessed not only for nominal error guarantees but for feasibility, semantic granularity, and adversarial robustness.

I. Introduction

A network intrusion detector turns traffic into scores, but a security operations centre (SOC) cannot act on a score; what it needs to know is how much of its alert queue is false. The standard evaluation vocabulary — AUROC, AUPRC, F_1 , recall at a threshold — measures how well a detector ranks and says nothing about how many alerts an analyst is handed will be real. That quantity is the

false discovery proportion, and its importance for intrusion detection has been understood since Axelsson’s base-rate analysis [1], [2]: at realistic prevalence a detector with excellent ranking still produces alert queues that are mostly noise.

Online multiple-testing procedures address this directly. Writing V_t for false discoveries and R_t for discoveries after t hypotheses, $\text{FDP}_t = V_t/(R_t \vee 1)$ and $\text{FDR}_t = \mathbb{E}[\text{FDP}_t]$; procedures such as LOND, LORD++, SAFFRON, ADDIS and their e-value counterparts [3], [4], [5], [6], [7], [8] control FDR over a stream, deciding each hypothesis before the next arrives. Conformal calibration [9], [10] supplies the evidence, distribution-free provided calibration and test scores are exchangeable, and e-values make the construction robust to the arbitrary dependence network traffic exhibits [11], [12]. Composing these off-the-shelf components gives what we call a statistical trust layer,

detector $\rightarrow$ evidence $\rightarrow$ aggregation $\rightarrow$ controller $\rightarrow$ alert.

The literature establishes that the constituent procedures are valid under their assumptions; composing them into this grouped pipeline raises separate questions of feasibility and adversarial exposure, and even the composed validity rests on a group-evidence premise we make explicit (Assumption 1 and Lemma 1). Validity does not establish usability: a valid guarantee is useful only if the evidence the detector can produce can cross the rejection threshold the procedure sets — and conformal evidence from a finite calibration set is bounded above, while online procedures shrink their level over time. Whether the first can exceed the reciprocal of the second over a security-realistic stream is answered by no validity theorem; we show it is often answered in the negative. Two questions follow: what the layer costs in the units a SOC works in, and what an adversary who knows it is present can do to it.

Contributions. We do not propose a new procedure, detector, or conformal construction; we characterise when the existing composition works, what it costs, and how it breaks. Two results carry the paper — a feasibility characterisation and an attackability result — joined by a granularity–resolution exchange rate that is their bridge rather than a third finding.

C1: a feasibility boundary (section IV). Take a procedure whose level is either multiplicative in a decaying spending sequence — $\gamma_t t^d \rightarrow 0$, a condition on γ that

must be checked per procedure and that a level growing quadratically in the rejection count would fail — or a lag-sum over rejection times, which needs no such condition. Under finite-resolution conformal evidence and an uninterrupted controller, the set of times at which it can reject during a rejection-free run is then finite, and the infeasible state absorbing: feasibility over a horizon T requires $|\mathcal{C}| \geq k/\alpha_T - 1$, which under the max-min-optimal horizon-uniform sequence is $kT/c_0 - 1$ in a cold-start coefficient c_0 (w_0 for LORD++, $2w_0$ for LOND/e-LOND). Two procedures escape, by distinct mechanisms and each at a measured price; so does removing either premise — smoothing the evidence or restarting the controller.

What is already known, and what C1 adds. That finite conformal resolution can leave online testing underpowered has been observed: Huo et al. [13] report α -death empirically for LOND, SAFFRON and ADDIS run on conformal evidence, easing as the calibration set grows, and respond with a different, local-FDR selection rule rather than an analysis of the composition (section XI). We characterise when that power loss becomes a structural impossibility: for two families covering LOND, LORD++, e-LOND and e-LORD — the e-value procedures as well as the p -value ones — we derive a finite absorbing discovery horizon, the exact calibration–horizon exchange rate, which procedures escape it and by which mechanism, and the design tradeoffs this forces at security-scale horizons.

The bridge: the granularity–feasibility exchange rate (section V). Coarsening the alerting unit buys feasibility exactly, because the required calibration size is linear in the hypothesis count in every grouping family we test. It costs resolution: against a fixed 5-minute atomic evaluation reference, each issued alert blurs together an increasing number of atomic units as the alerting unit coarsens (from one to tens; section V). This is not a third result — under uninterrupted deployment-wide control for the covered families, aggregation is the direct route to feasibility, and the aggregation it forces is exactly what C2 attacks. (Malicious-flow mass is not a second reading of that cost: dominated by which episodes fire, it is reported per configuration while we rest the resolution claim on alert blur.)

C2: the trust layer as an attack surface (section VI). There are two surfaces, one within a hypothesis and one across them. Surface A is a low-footprint evasion: no symmetric e-merging family that can fire resists padding, and within the pre-committed position-weighted class we analyse, abandoning symmetry converts appended padding into a strictly cheaper front-loading attack rather than removing it. We do not claim this for asymmetric merging in general. Surface B shows the controller’s adaptive state is attacker-controllable: one of C1’s escape mechanisms, exemplified by ADDIS, conditions its spending index on observed evidence and so exposes that index in closed form, letting an attacker silence it permanently — conspicuous in volume rather than cheap. Because the only stream

on which ADDIS both escapes the horizon and carries a valid guarantee is a synthetic one (the two properties are inseparable on two-point conformal evidence), the real-stream demonstration necessarily sits at the known-invalid window, and we reproduce the mechanism on a guarantee-valid synthetic stream to show it is not an artefact of that (section VI-F). The other escape, online e-BH’s history-wide fixed point, is different in kind, and we do not demonstrate this attack against it. Where both surfaces apply to the same procedure, they coincide.

II. Background and System Model

A. Pipeline and conformal evidence

Figure 1 shows the composed system. Flows $x_1, x_2, \dots$ arrive in timestamp order; a detector f scores each $s_t = f(x_t)$, and a calibration set $\mathcal{C}$ of scored benign flows, disjoint from training and preceding deployment, converts scores into evidence. With K_t the rank of s_t among the calibration scores from the top, the threshold conformal e-value is $e_t = \frac{|\mathcal{C}|+1}{k} \mathbf{1}\{K_t \leq k\}$, with $\mathbb{E}[e_t] \leq 1$ under the null and $p_t = \min(1, 1/e_t)$. It is two-point — a flow either beats the k -th largest calibration score or not — so the evidence is bounded, by

$$M = \frac{|\mathcal{C}| + 1}{k} \quad (1)$$

No flow, however anomalous, exceeds eq. (1). This is the finite resolution of deterministic rank-based conformal evidence against a finite calibration set [9], [14] — the unsmoothed threshold construction considered here, not a feature of our detector — and the quantity the rest of the paper turns on. Randomising the rank (section IV-D) removes the floor, at a measured price.

The calibration premise. Every guarantee assumes $\mathcal{C}$ contains no attack traffic, and section IX-C measures its worth: one flow in the adversarial worst case, since at $k = 1$ the rule is a single order statistic. The tolerable quantity is a count, not a rate — and adversarially it is zero.

B. Security-semantic grouping

A detector predicts on flows; a SOC acts on something larger. We group flows into units G_j and aggregate their evidence into one quantity $\text{Ev}(G_j)$ per unit, the hypothesis the online procedure tests. Since LSPR23 provides no usable flow-to-campaign ground truth, the deployable units here are episodes — host-pair–time-bucket groups — never incidents or campaigns. Our aggregation is the arithmetic mean $\text{Ev}(G) = \frac{1}{m} \sum_{i=1}^m e_i$, valid at every arity, so every procedure comparison and both attacks use this uncapped mean; a pre-committed cap $\sum e/n_0$ (valid only for $m \leq n_0$) enters only as a competing policy (section IX-D). The step to a valid group e-value is algebraic for a fixed collection, but here the group is metadata-selected with random arity, and different controllers need different things of it.

What each controller requires. e-LOND and online e-BH control FDR under arbitrary dependence provided

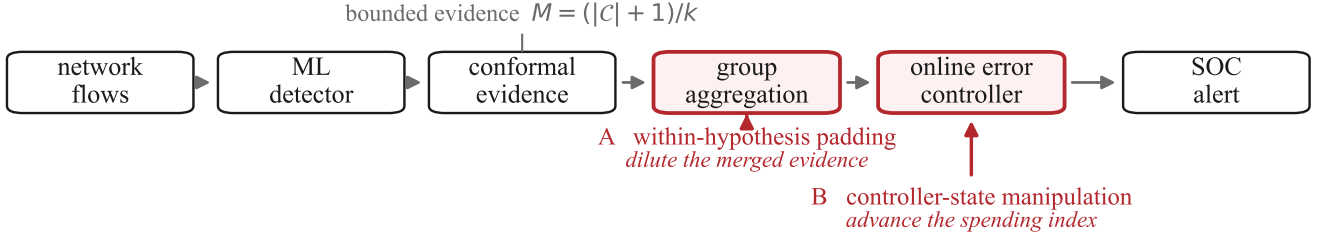


Fig. 1. The statistical trust layer, and the two places an adversary reaches it. Evidence entering the controller is bounded by $M = (|C| + 1)/k$, which section IV shows creates a finite discovery horizon. Surface A attacks a hypothesis’s composition: the grouping key is made of fields the adversary chooses, so it adds ordinary flows to its own episode, diluting the merged evidence (section VI-C). Surface B attacks the controller’s sequential state: a procedure escaping the horizon by conditioning its spending index on observed evidence exposes that index to whoever generates the evidence (section VI-F). Neither requires model or detector access, or any perturbation of the attacker’s own flows.

each true-null group carries a marginally valid e-value, $\mathbb{E}[\text{Ev}(G_j)] \leq 1$ — weaker than the conditions the p-value procedures need (conditional super-uniformity, independence, monotonicity; ADDIS additionally uniform conservativeness, which section IV-C shows it lacks here). The guarantee-bearing claims therefore use e-LOND, and the property to establish is group-level marginal e-validity.

The conditioning must be done per hypothesis, not globally, or the attack analysis below does not go through: a premise stated against one global metadata σ -field says nothing once the adversary enriches that σ -field elsewhere in the stream, because conditioning on more information can raise a conditional expectation. So for each group j let $\mathcal{M}_j$ be the σ -field generated by the pre-committed metadata that determines that hypothesis and nothing else — the grouping key, the membership indicators $\mathbf{1}\{i \in G_j\}$ and arity m_j of G_j , its position in the pre-committed order (hence the spending weight γ_j it is offered), and the horizon T . It carries no information about any other group’s composition. Note γ_j , not the level: for e-LOND $\alpha_t = \alpha\gamma_t(R_{t-1} + 1)$ depends on realised past rejections, which is exactly what the procedure’s own proof permits. “Pre-committed” means chosen before the scores and not score-adaptive, not statistically independent of them.

Assumption 1 (Metadata-conditional per-flow validity). For every benign constituent flow i of a true-null group j , the per-flow e-value satisfies $\mathbb{E}[e_i | \mathcal{M}_j] \leq 1$; equivalently, for the two-point conformal e-value, $\Pr(K_i \leq k | \mathcal{M}_j) \leq k/(|C| + 1)$. This is a sufficient condition, stronger than the unconditional $\mathbb{E}[e_i] \leq 1$ that split conformal supplies (and weaker than the same requirement against a global metadata σ -field); it holds when the benign score is not adaptive to that group’s metadata — e.g. conditional exchangeability of calibration and null deployment scores given $\mathcal{M}_j$ — a genuine modelling assumption, since endpoint or service metadata can correlate with the timing and volume features that drive the score.

Lemma 1 (Group-level marginal e-validity). Under Assumption 1, every true-null group with $m_j > 0$ has a

marginally valid group e-value, $\mathbb{E}[\text{Ev}(G_j)] \leq 1$, for random membership and random arity m_j and under arbitrary dependence among the e_i . The group-level stream is therefore marginally e-valid, exactly what e-LOND and online e-BH require.

Conditioning on $\mathcal{M}_j$, membership and m_j are fixed, so $\mathbb{E}[\text{Ev}(G_j) | \mathcal{M}_j] = \frac{1}{m_j} \sum_{i \in G_j} \mathbb{E}[e_i | \mathcal{M}_j] \leq 1$, and the tower property removes the conditioning (appendix A). Within-group dependence is irrelevant (only linearity is used) and between-group dependence arbitrary (e-LOND’s proof needs only marginal validity).

Why not calibrate the group directly? The obvious way to make this hold by construction is to stop merging per-flow evidence and run split conformal on the grouped unit, with the same protocol on the calibration split; the premise would then be exchangeability of calibration and benign test groups, with no per-flow conditional statement. We build it, and it is worse in three ways at once (section IX-B): the premise is discharged but replaced by ordinary group exchangeability, which arity heterogeneity suggests may be fragile under group-formation shift; the evidence ceiling falls by the flows-per-group ratio; and the alerting stops entirely at the primary and replication windows. The last of these needs no view on the premise at all. Assumption 1 is what survives when feasibility is a requirement rather than an afterthought.

Two clarifications keep this honest. First, “marginal” is over the test scores and the calibration draw, not conditional on the realised calibration set; the Bates calibration-conditional adjustment (table XXXVIII) is stronger than e-LOND needs, so we treat it as optional. Second, the attacked group is a false null, and Assumption 1 — like e-LOND’s FDR proof — constrains only true-null groups. Padding changes the attacked non-null group’s membership and arity (Theorem 10), but the pads carry its own key, so under a key-determined order they add no hypothesis and leave every true-null group’s $\mathcal{M}_j$ unchanged, so Assumption 1 holds verbatim on the attacked stream. This is a second, validity, reason to fix the within-bucket order by group metadata rather than by first-flow arrival (section IX-D): an arrival-ordered tie-

break lets a pad shift a true-null group’s position, hence its γ_j and $\mathcal{M}_j$. The attack therefore exploits a rule valid under the premise rather than violating it. Where the premise fails — section IX-A finds the one window (0.85) whose measured benign firing rate gives $\mathbb{E}[e] \approx 51$, refuting even marginal e-validity — no FDR guarantee attaches, and every number there is a measurement against labels. That firing-rate check is a diagnostic: it can refute validity, but a compatible count does not prove it.

C. Online procedures and metrics

Each procedure offers a level α_t at step t and rejects when the evidence exceeds $1/\alpha_t$. What distinguishes them, for our purposes, is not their FDR proof but what index advances the spending sequence. Table III maps the procedures we test on that axis. It is the map that predicts which of them are covered by Proposition 2 and Proposition 3 and which escape.

Metrics. We report FDP and FDR; malicious-flow coverage and episode recall, whose divergence is the subject of section V; alerts issued; and one feasibility quantity,

$$\text{margin} = \frac{(|\mathcal{C}| + 1) w_0}{T} - 1, \quad (2)$$

the horizon-uniform feasibility slack of a level- w_0 procedure such as LORD++ (LOND/e-LOND fire for a band where eq. (2) is mildly negative, their first-step level being $\alpha = 2w_0$).

The terms the paper holds distinct — validity, feasibility, detection power, episode recall, atomic coverage, alert blur, and the two attack costs — are collected in table II.

Conventions for the results. Four hold throughout, so we state them once rather than at each number. First, the per-alert padding cost r^* is an oracle bound (it uses the realised evidence sum and the unperturbed trajectory) and a lower bound on what an attacker must spend; the paper’s other attack quantities are not, and each says which it is where it appears — G^* and B^* are exact closed forms verified by replay, the keyed-hash $N_{50/90/99}$ are empirical reliability budgets over 400 draws, and the front-load L^* and the whole-detected-set padding totals are upper bounds. Second, every headline absolute result — table I and the operational evaluation — is reported under the canonical within-bucket order, a hash of the group’s own metadata key, neither attacker-timed nor detector-favourable (section IX-D). Paired ablations and sensitivity analyses may instead run under the shipped first-flow arrival order, the optimistic upper bound over the orders we audit, and every table and figure that does so says so in its caption. Where a result is a contrast between two procedures on one stream (smoothing, restart, weighting, controllers) both legs use the same order, so the order shifts the level and not the sign. Third, position 0.55 is the primary analysis window and 0.62 the replication window — pre-committed as the analysis pair before any tail diagnostic existed, not selected by one — while 0.85

is a known-invalid stress-test window whose evidence is not a valid e-value and which we use only for mechanism and attack-cost measurement (section VII). Every FDR statement at every window is conditional on Assumption 1: the benign-tail diagnostic refutes 0.85 and cannot establish the other four. Fourth, the feasibility margin depends on $|\mathcal{C}|$, k and T alone, never on the scores.

III. Threat Model

Defender. Controls the detector, calibration set, grouping rule, evidence construction and controller, all correctly implemented — no attack relies on an implementation flaw. Adversary goal. Suppress the alert on its own activity; secondarily, delay it or silence the controller for every later hypothesis.

Capability. The adversary generates network flows from compromised or attacker-controlled hosts. Because the grouping key is (SrcIP, DstIP, time bucket), it controls group membership by construction: ordinary connections to the host it is already attacking, in the same bucket, land in the same group — needing no flow crafting, perturbation, or detector access. It may also create precursor episodes and order its own flows.

Knowledge. White-box: k , $|\mathcal{C}|$, procedure parameters and controller state. Grey-box: procedure and parameters but not $|\mathcal{C}|$ nor the state — for section VI-F a $\pm 33\%$ estimate of $|\mathcal{C}|$ suffices and unknown state costs $\approx 36\times$ over-provisioning. Black-box: no detector access; ordinary traffic of a chosen volume on a common service.

Cost metric. Padding flows emitted, converted to bytes and rate in section VI-F; where estimated from realised detector output it is a lower bound.

Out of scope. We exclude poisoning the detector’s training set, compromising the SOC, arbitrary modification of stored scores, and any claim that these attacks have been seen in the wild. Calibration-set contamination is in scope (section IX-C): it needs only that a labeller miss an attack flow.

IV. When Can an Online Guarantee Produce Any Alert?

Known vs. established here. That bounded evidence loses to a shrinking threshold is elementary and “ α -death” is known, held cured by α -investing [15]. We establish the exact structural conditions under which procedures fall silent, that the state is absorbing with a cold-start deadline, the scaling $|\mathcal{C}| \geq kT/c_0 - 1$, and that α -investing does not cure it here.

Three feasibility notions, kept distinct. A step t is evidence-feasible if the largest admissible evidence can cross the offered threshold ($\alpha_t \geq 1/M$); a procedure is cold-start feasible through T if a rejection is possible at every $t \leq T$ on a rejection-free prefix; a state is absorbing if, once entered on such a trajectory, no later observation can produce a rejection.

A. A finite discovery horizon

The argument is elementary. Evidence is bounded by M and a rejection at t requires $E_t \geq 1/\alpha_t$, so a rejection is possible at all only if

$$\alpha_t \geq \frac{1}{M}. \quad (3)$$

Every procedure in the two families below pays its level from a finite budget, so during a rejection-free run α_t cannot recover: it decays under a decaying γ , and is pinned at the constant cold-start value c_0/T under horizon-uniform γ . Equation (3) therefore has a last time it holds, after which the procedure is silent — no observation can change its output. The two families cover four of those we test (LOND, e-LOND, LORD++, e-LORD); SAFFRON sits outside their letter yet does not escape in practice (table III), and the two genuine escapes — ADDIS and online e-BH — are located in section IV-C: finding them is part of the result.

Proposition 2 (Multiplicative family). Let $\alpha_t = \alpha \gamma_t g(\mathcal{H}_{t-1})$ with $\gamma \geq 0$, $\sum_t \gamma_t \leq 1$, and $g \leq c(R_{t-1} + 1)^d$ for the rejection count R . Since $R_{t-1} \leq t - 1$, a rejection is possible at t only if $\gamma_t t^d \geq 1/(\alpha c M)$. If $\gamma_t t^d \rightarrow 0$, the set of such t is finite; if $\gamma_t t^d$ is additionally eventually non-increasing, feasibility ends at a last time after which no later step is feasible, and the infeasible state is absorbing.

The degree condition is asymptotic and must be checked per procedure. For the horizon-free $\gamma_j \propto j^{-1.6}$, $\max_t t\gamma_t = 0.4375$ (at $t = 1$) with $t\gamma_t \rightarrow 0$, so $d = 1$ holds; a procedure whose level grew quadratically in the rejection count would escape the argument. The horizon-uniform $\gamma_t = 1/T$ falls outside the letter of the condition — $t\gamma_t = t/T$ rises to 1 — but needs none: before any rejection the level is the constant c_0/T , so feasibility over the horizon is all-or-nothing and Corollary 4 decides it, while Proposition 3, which assumes no monotonicity of γ , covers the horizon-uniform case unchanged.

Proposition 3 (Lag-sum family). Let $\alpha_t = w_0 \gamma_t + \sum_j a_j \gamma_{t-\tau_j}$ over rejection times $\tau_j < t$ with $0 \leq a_j \leq \alpha$ for every j (LORD++ has $a_1 = \alpha - w_0$ and $a_j = \alpha$ for $j \geq 2$). During a rejection-free run of length Δ every lag is distinct and at least Δ , so, using $\gamma_{t-\tau_j} \leq S(\Delta)$ and $a_j \leq \alpha$ term by term, $\alpha_t \leq (w_0 + \alpha)S(\Delta)$ with $S(\Delta) = \sum_{d \geq \Delta} \gamma_d$. Since S is non-increasing and $\sum \gamma < \infty$ implies $S(\Delta) \rightarrow 0$, the quantity $\Delta^* = \min\{\Delta : (w_0 + \alpha)S(\Delta) < 1/M\}$ is finite, and because Δ only grows during a rejection-free run the infeasible state is absorbing — with no monotonicity assumption on γ itself.

Proposition 2 covers LOND, e-LOND and e-LORD (the last via the equivalence e-LORD = e-LOND under $\gamma_t = \omega_t \prod_{j < t} (1 - \omega_j)$ [16]), and Proposition 3 covers LORD++. We do not extend the bound to the remaining e-GAI members — e-SAFFRON at $\lambda = 0.1$ is a testing level the equivalence does not cover, and mem-e-LORD

controls a different error rate. Both propositions are verified numerically. The result is a precise scope, not a claim that online FDR “dies”.

B. Horizon-uniform feasibility bound

Corollary 4. Feasibility over a horizon T with no prior rejections requires $|\mathcal{C}| \geq k/\alpha_T - 1$. Write c_0 for the cold-start coefficient — the constant multiplying γ_T in the level a procedure offers before its first rejection, so $c_0 = w_0$ for a level- w_0 procedure such as LORD++ and $c_0 = \alpha = 2w_0$ for LOND/e-LOND. Under horizon-uniform $\gamma_t = 1/T$, which is max-min optimal, this is

$$|\mathcal{C}| \geq \frac{kT}{c_0} - 1.$$

Corollary 5 (Calibration-horizon ratio). Coarsening or refining the alerting unit moves $|\mathcal{C}|$ and T together. If a construction replaces a calibration set of $|\mathcal{C}|$ units by one of $|\mathcal{C}'|$ units while leaving the horizon at T' , it is feasible only if $|\mathcal{C}'| \geq kT'/c_0 - 1$. The requirement is therefore a scale-free count ratio k/c_0 — 20 at $k = 1$, $c_0 = \alpha = 0.05$ — independent of the dataset: a deployment needs 20 calibration units per hypothesis it intends to test, whatever the unit is.

This is the most portable statement of C1, because a practitioner budgets in it. Its calendar form is not scale-free: converting 20 units per hypothesis into observation time multiplies it by the rate at which the calibration and deployment windows produce units, so an attack-free observation period of 8.5–47 $\times$ the deployment span here (through a measured group-rate ratio of 0.42–2.35) is a property of this stream, while the factor 20 is not. Section IX-B is Corollary 5 applied: calibrating the grouped unit directly replaces 1.8–2.4M flows by 37–57k groups, and 20 groups per hypothesis is exactly what the construction cannot supply.

Figure 2 plots this against the horizon. At $k = 1$, $w_0 = 0.025$ it is 6.5×10^8 calibration flows at LSPR23 scale for $c_0 = w_0$ (3.3×10^8 for LOND/e-LOND, $c_0 = \alpha$), 1.44×10^9 for one hour of a 10k-flows/s link, 3.46×10^{10} for one day, and 3.2×10^{13} under horizon-free $\gamma \propto j^{-1.6}$. The windows we evaluate carry $|\mathcal{C}|$ between 1.81M and 2.45M. The same algebra appears in Krönert et al. [17] as $|\mathcal{C}| = \nu m / \alpha - 1$, answering a different question about a windowed procedure’s exactness rather than uninterrupted feasibility (section XI).

The squeeze is sharpest at the very start of the stream. Before its first rejection a procedure carries no history term, so it has only a short cold-start window in which a rejection is possible at all: under $\gamma \propto j^{-1.6}$, the first 18 hypotheses at $|\mathcal{C}| = 10^4$ and 334 at 10^6 (29 and 515 for LOND/e-LOND), after which the absorbing state of Propositions 2 and 3 is entered. At the $\pi = 10^{-4}$ low-prevalence sensitivity point that window almost always closes empty. The first true attack is not expected until index $\approx 10,000$, so a rejection arrives in time to bootstrap

the controller with probability only $\approx 0.3\%$; the sole alternative, an early false rejection, has probability at most a few percent (4.5% for e-LOND, 2.2% for LORD++, in simulation at a stated $|\mathcal{C}|$; the benign-only arm of table XXXIV produces no rejection on any stream, but with five effective windows that is consistent with this rate rather than a test of it, $\Pr(\text{zero} \mid 4.5\%) = 0.79$). The published α -death fix does not help, being identical to the base procedure until that first rejection and weaker after (table III).

And live-fire prevalence is the optimistic case. LSPR23 is an exercise: 10.06% of flows and 0.48–0.81% of episodes are malicious, so we sweep prevalence as a parameter down to $\pi = 10^{-5}$ and to zero — two to three orders below the exercise, and a sensitivity analysis rather than a claim about any particular SOC’s base rate. Thinning malicious episodes to a target rate — replacing each with benign evidence from the same window, so T , the margin and every survivor’s step index are held exactly fixed — separates the two quantities cleanly: feasibility does not move at all, while at $\pi = 10^{-3}$ e-LOND’s mean detections fall from 0–72 to 0–8 and at $\pi = 10^{-4}$ its probability of ever rejecting is 0–40%, mostly under 10% (table XXXIV). Deleting the hypotheses instead — which shortens T and raises the margin by up to +0.013 — gives the same collapse to within 0.02 in that probability, so it is prevalence and not re-indexing that does the work — a distinction worth drawing, because deletion provably can help: removing a hypothesis shifts every later one into a higher α_t , so a survivor that could not clear its own step may clear its new one. The two escapes collapse with it. Every detection count in this paper is therefore an upper bound in one more respect, and the separation the paper draws between feasibility and detection is visible here driven by base rate alone: the margin improves while detection goes to zero. This is a sensitivity, not a full counterfactual — a genuinely low-prevalence deployment would also have a different detector, calibration corpus and benign traffic — and e-TOAD’s collapse is measured over fewer draws than the other two, so it is directional.

C. Which procedures escape, and what the escapes cost

Two procedures escape, and among the procedures studied here the boundary is sharp: the horizon binds a procedure whose spending index advances on every hypothesis, and is escaped by one that advances it only on tested hypotheses or defers the decision. This is a classification of what we examined, not a characterisation of every possible procedure — Proposition 2 leaves room for a third route, escape by history-dependent level growth fast enough to outrun γ_t , which none of these takes. ADDIS takes the first route. Its index counts hypotheses that are selected ($p \leq \tau$) but not candidates ($p \leq \lambda$), and at $k = 1$ no episode p-value ever lands in $(\lambda, \tau]$, so the index never advances, the level never decays, and ADDIS is silent 0.0% of the time. But this voids the guarantee:

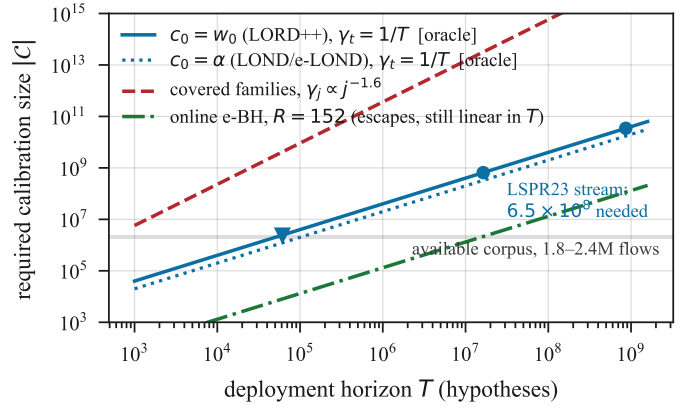


Fig. 2. How large must the calibration set be to keep a rejection possible over a deployment horizon? At $k = 1$, $w_0 = 0.025$ the requirement $kT/c_0 - 1$ is linear in T for every procedure covered by Propositions 2 and 3, differing only through c_0 (a factor 2 between LORD++ and LOND/e-LOND). Online e-BH is plotted for comparison and is not covered — it escapes the absorbing horizon (section IV-C) — but still carries a calibration requirement, linear in T per simultaneous discovery. Both outrun the 1.8–2.4M calibration flows these windows supply by orders of magnitude.

almost no null is selected (5 of 31,313, first-flow), so its realised FDP is only what the labels say, and under the max-min-optimal sequence the wealth discount drops the initial level below the conformal floor, silencing it outright. Online e-BH takes the second: its step-up threshold is a fixed point over the whole history [8], [12], [18], so a rejection-free prefix is simply not absorbing. It still needs the calibration budget (R simultaneous discoveries require $|\mathcal{C}| \geq T/(\alpha R) - 1$), and its advantage is robustness to a misspecified horizon: at the stress window and one seed — the only cell we sweep the horizon on, so read it as an illustration of the mechanism rather than a measured rate — online e-BH keeps 144 of its 152 rejections under a $100\times$ over-estimate, where LOND keeps all 151 at $2\times$ but none at $100\times$, and LORD++ none already at $2\times$.

The second route is not one procedure but a one-parameter family, and the parameter is alerting latency. Under decision deadlines [19] each hypothesis must be decided by a time $d_t \geq t$ or accepted for good; $d_t = t$ recovers e-LOND and $d_t = \infty$ recovers online e-BH [20], so the two ends of the boundary above are joined by a continuum. Its operational reading is direct — d_t is how long a SOC will let a hypothesis sit undecided — and a bucket-close deadline is exactly the batched architecture a SOC receives, arbitrary-dependence valid where a batched BH is not (section XI). Deferring does not buy feasibility. Measured exactly, the fraction of hypotheses that cannot be rejected at their own arrival is unchanged by the deadline — 96.3% at the primary window whether decisions are immediate, held to bucket close, or never forced (90.1% under first-flow arrival; table XIII) — because reaching a step-up count of R needs R hypotheses to clear the bar at once, and on our streams at most 0.48% of episodes

carry any evidence at all. The escape works by revisiting, so it is paid in latency — and under the order table I reports it buys almost nothing. Across the ten window-seed cells, deferring to bucket close converts to power at exactly one, and there it buys one detection (34 → 35 at 0.85), the known-invalid window whose evidence is not a valid e-value anyway. Under first-flow arrival — the optimistic bound — it converts at two cells (13 → 32 at 0.62, 30 → 31 at 0.70) and at none of the other eight. Latency is a lever on the controller, not on the evidence.

Do the newest procedures escape the cold-start barrier? No. The strict improvements over e-LOND published under arbitrary dependence — a closure construction and a compound-e “donation” construction, two recent strict improvements directly applicable to our setting [20], [21] — fail to reach a first rejection before it, for two different reasons.

Proposition 6 (Donation is bounded before the first rejection). For donation e-LOND with parameter $\delta \in (0, 1]$, on a rejection-free prefix, $\alpha_t \leq \delta\gamma_t/(1 - \delta)$. Hence donation e-LOND satisfies Proposition 2 with $c = 1/(1 - \delta)$, $d = 1$: the horizon stays finite and the infeasible state absorbing.

The mechanism is one line: donation’s wealth is γ -weighted and $\sum \gamma \leq 1$, so the donation budget is the spending budget, and redistributing wealth cannot manufacture it. At $\delta = \alpha$ the cold-start coefficient moves only from α to $\alpha/(1 - \alpha)$ — the calibration requirement shrinks by a factor $(1 - \alpha)$, i.e. 5%, against a shortfall of $20\times$ (Corollary 5). What Proposition 6 establishes is therefore precisely this: donation cannot rescue a controller that fails to obtain its first rejection before the bounded-evidence cold-start horizon. It is not a global cap and must not be quoted as one — after $1/\delta$ rejections the denominator can vanish and the level is not bounded by this expression at all — and it is sharp where it applies. Runs here do pass that count — 30, 31 and 73 rejections at 0.70, 0.77 and 0.85, where the realised boost reaches 1.83 against the cold-start ceiling 1.05. What we observe is the weaker and sufficient fact that the denominator never actually vanishes on these streams, the realised donation wealth staying at most 0.12 of a budget of 1.

Proposition 7 (Closure is bounded by its zero-evidence subset). For closed e-LOND, let $Z_t = \#\{i < t : E_i = 0\}$. Then $\alpha_t \leq \delta\gamma_{Z_t+1}(R_{t-1} + 1)$.

Corollary 8 (When the closure is absorbing). If along a rejection-free run $Z_t \rightarrow \infty$ fast enough that $\gamma_{Z_t+1} \rightarrow 0$, closed e-LOND enters the absorbing infeasible state. Two sufficient conditions, one finite and one asymptotic. Let $H = \min\{j : \gamma_j < 1/(\delta M)\}$ be Proposition 2’s index horizon. (i) With $P_T = \#\{i \leq T : E_i > 0\}$ the number of hypotheses carrying any evidence, the run is absorbed by step $H + P_T$ whenever $H + P_T \leq T$ — an additive offset, holding pathwise along the rejection-free run with no assumption beyond counting. (ii) Asymptotically, if

$\liminf_t Z_t/t > 0$ then for every $\rho' < \liminf_t Z_t/t$ the horizon is eventually Proposition 2’s rescaled by ρ' .

This is a genuinely new covered case rather than an instance of Proposition 2 — the closure’s level is not γ_t times a bounded factor — so C1’s scope widens rather than merely extending. The zero-evidence hypotheses form an admissible subset, giving the same decay under a rescaled index. The bound is unconditional; the conclusion is not, and Corollary 8 carries its hypothesis. Form (i) is the one that applies here, and it needs no density argument: a stream carries P_T hypotheses with any evidence at all, and the closure cannot outrun its own index by more than that many steps. Our streams make the offset small — P_T is 0 to 152 across the ten window-seed cells, at most 0.48% of T — so the closure’s horizon exceeds Proposition 2’s by at most 152 hypotheses and the cap is attained. Where instead most hypotheses carry positive evidence P_T approaches T , Z_t stays small and the bound says little: that is the regime the closure was designed for, and it is not the sparse-evidence regime studied here. Both proofs are in appendix A. The r-LOND variants inherit the classification verbatim: their rejection sets coincide with the e-LOND ones on calibrated p -values.

Measured, under the canonical order the two buy +0 and +0 true detections across ten window-seed configurations, and +1 and +0 under first-flow arrival (table XIII), because these streams have almost no evidence to redistribute. The barrier is a property of the evidence, not of the controller’s sophistication. For e-TOAD we use the interpretation of the active set and the step-up upper limit that is consistent with the stated limiting cases; the alternate literal interpretation is included as a sensitivity analysis and does not affect our conclusion (table XIII).

D. Removing the premises: smoothing and restart

Both premises are load-bearing, and removing either carries a measured cost (appendix E). Randomised smoothing replaces the discrete p -value with a $\text{Unif}(0, 1)$ draw [9], [14] and dissolves the absorbing state — but the extra detections are a lottery. Both legs of this contrast, and of the restart and weighting contrasts below, run under first-flow arrival — the order shifts the level an alert fires at, not the sign of a paired comparison (section IX-D). At the primary window smoothed LOND rejects 24.1 ± 2.7 episodes against 18 discrete, yet the set detected reliably (probability ≥ 0.9) is still those same 18, and the surplus does not survive arbitrary-dependence-valid merging (Hommel gives 12.5). Restart instead buys back real power, and its design chooses the cost: resetting the budget each epoch raises recall from 0.065 to 0.345 but forfeits the deployment-wide guarantee ($\text{FDR} = 1 - (1 - q)^n$ grows with the epoch count), while pre-allocating it ($\sum_i q_i \leq q$) keeps deployment-wide $\text{FDR} \leq q$ at some of that power (recall 0.195; table XII). Restart forces a power-versus-guarantee choice, then, rather than voiding the guarantee.

TABLE I

Main cross-window results, detector seed 0 (the ten-row two-seed matrix is in appendix D). Two-hour host-pair grouping, e-LOND, $\gamma \propto j^{-1.6}$, $k = 1$, $q = 0.05$; margin is eq. (2). “Detected” is e-LOND true-positive rejections under the canonical metadata-hash order, with the shipped first-flow order — an optimistic upper bound over the orders we audit — beside it (section IX-D); “median pad” is the black-box suppression cost of the canonical detections against the running level $1/\alpha_t$, a lower bound. 0.55 is the primary analysis window, 0.62 the replication window — pre-committed, with every FDR statement at all five conditional on Assumption 1 and Lemma 1. 0.85 is the known-invalid stress-test window, its benign tail an unambiguous violation ($\mathbb{E}[e] \approx 51$); the other four are merely not refuted by a diagnostic resting on 0–3 benign firings (table XXX), and the metadata-stratified reading is not clean at any (table XXXI). Position 0.70 carries the feasibility-is-not-detection separation: at seed 1 the same +0.536 margin with AUROC 0.914 $\rightarrow$ 0.847 takes first-flow detections from 30 to 0, and the canonical order gives 0 at both seeds. The canonical column is one draw from the 50-order ensemble (medians 1, 6, 2.5, 1, 23.5; maxima 10, 11, 12, 6, 44), and not a uniformly typical one: it sits above the ensemble median at 0.55, 0.62 and 0.85 — at the maximum at 0.62 — and below it at 0.70 and 0.77. The first-flow column exceeds every ensemble maximum (table X).

Position	AUROC	T (episodes)	$ C $	margin	malicious ep.	canonical order detected	median pad	first-flow (upper bd.)	tail vs. nominal
0.55	0.916	57,368	2,448,993	+0.067	275	3	24	18	compatible (1.07 $\times$)
0.62	0.799	49,267	2,449,031	+0.243	398	11	6	13	compatible (1.16 $\times$)
0.70	0.914	37,231	2,287,988	+0.536	287	0	—	30	compatible (3.79 $\times$)
0.77	0.828	31,672	2,116,418	+0.671	246	0	—	31	compatible (2.72 $\times$)
0.85	0.999	31,568	1,813,113	+0.436	255	34	116	72	violated (50.9 $\times$)

V. The Granularity–Guarantee Tradeoff

Corollary 4 makes the exchange rate explicit: since $|C|_{\min} = kT/c_0 - 1$ in the cold-start coefficient c_0 ($= \alpha$ for the e-LOND controller we report), grouping buys feasibility directly by reducing T from T_{flows} to T_{episodes} — an exact lockstep, not a heuristic improvement. We test five families a SOC would recognise (source–dest host pair, source host, dest host, /24 subnet pair, source–service), each over six bucket widths from five minutes to one day plus a whole-deployment bucket — 35 configurations in all (appendix D).

A. The empirical granularity curve

Two things happen as the unit coarsens (fig. 3).

Feasibility is bought exactly. The required $|C|$ is linear in T in every family, so each becomes feasible once the bucket is coarse enough. Source-host grouping is feasible at every bucket, collapsing each window to 763–24,043 episodes and needing well under the 1.8–2.4M calibration flows available — and an attacker-host alert is an ordinary SOC unit, so the barrier can be bought off.

Resolution is what pays. Against a fixed atomic reference — the 5-minute src-dst malicious episodes — each issued alert blurs from 1 of them to ~ 20 –40 as the bucket coarsens (fig. 3), localising the attack to a day-long host-pair blob. Episode recall (moving denominator) also falls under the oracle sequence (0.66 $\rightarrow$ 0.49 at 0.85), but its direction is γ -dependent, so we rest the cost on the blur.

That atom is our own unit, so we tested it against one we did not choose — and report what that test can and cannot support. LSPR23 ships the red team’s task record: 288 tasks, 295 timestamped step submissions, an attacker-defined decomposition of the same campaign. The tasks’ declared network segments are the dataset’s own per-flow seg vocabulary under a constant prefix, corroborated against the machine-readable compromise reports (83 agree, 0 disagree). Counting distinct red-team steps merged into each issued alert, the external count rises

with the alerting unit as the proxy does: at 0.70 it runs 4.3 $\rightarrow$ 27.0 across 5 min–24 h buckets against the proxy’s 1.0 $\rightarrow$ 28.3 (rank agreement +1.000 over six widths, +0.800 under the canonical order, +0.816 at 0.77). But we do not claim this as corroboration, for three measured reasons (table XXXIII). (i) A null that rotates the red-team timeline against the stream — preserving task identity, ordering and inter-arrival structure — reproduces the observed counts at 18 of 19 cells: the rise is driven by the bucket width capturing more submissions, not by alert-level correspondence. (ii) The record sees the alerts at only 3 of 10 window-order cells, and the reason is Proposition 2 again — issued alerts sit in the feasible prefix, the first minutes of a deployment window, and at 0.55 and 0.62 that predates the exercise’s first submission, so only 2% of alerts overlap it in time at all. An external record can only speak about an alert it coincides with, and the feasibility boundary puts the alerts where the record is silent. (iii) Every attributed alert sits on a labelled-malicious episode and the labels came from this same exercise, so nothing here is label-independent evidence that an alert is a true positive. Two further limits: the whole result rests on six coarse segment labels — across every row the compromise-IP conjunct admits no alert the segment conjunct did not — and at the 24 h bucket both nulls tie the observation because every submission shares one bucket, so only sub-daily widths inform. The honest reading is that the external unit agrees in direction and is too coarse to discriminate: the resolution cost survives the substitution but is not independently confirmed by it.

Malicious-flow coverage is not a clean measure of the resolution cost, which is why the exchange rate does not rest on it. It is dominated by the traffic mass of whichever episodes happen to fire — the top three host pairs carry 45–87% of malicious traffic — so it swings by orders of magnitude both across windows and with the bucket (table IX). We therefore report it per configuration and rest the resolution cost on the blur.

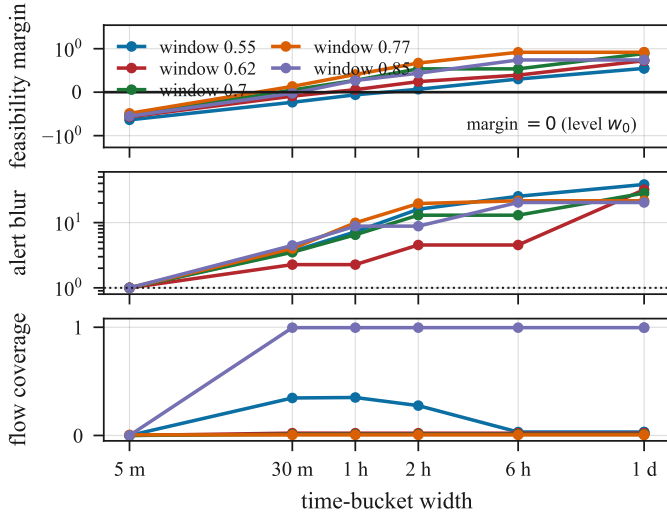


Fig. 3. Per-window granularity for the source-dest pair unit (five windows, seed 0). Coarsening the bucket buys feasibility (top: margin rises through zero) and costs resolution (middle: alert blur — fixed 5-minute reference units per alert — climbs from 1 to 20–40). Malicious-flow coverage (bottom) is not a clean resolution measure: set by whether the window’s dominant episodes fire, it varies across windows and with the bucket (table IX).

Coarsening is not the only lever: periodic restart (section IV-D) divides the hypothesis count a run must survive, preserving episode resolution (hourly restart at one-hour grouping reaches recall 0.406 at ≈ 125 alerts against 0.103 for six-hour coarsening). The tradeoff is granularity against the form of the deployment-wide guarantee; the exchange rate above concerns a single uninterrupted controller.

VI. The Statistical Trust Layer as an Attack Surface

Section V showed that aggregation is the direct route to feasibility. The catch is that both the aggregation and the adaptive state that lets a procedure escape Propositions 2 and 3 are reachable by an adversary who only generates traffic — no detector access, no crafted flows. Surface A is a practical, low-footprint evasion: suppressing the alert on one’s own episode costs tens of ordinary flows at the primary window (a median of 24 under the order we report, 118 under the detector-favourable one). Surface B is a structural controllability result about one of the escapes: an attacker who manufactures evidence advances a spending index that is conditioned on that evidence, and drives it to permanent failure. It is conspicuous in volume rather than cheap ($37.5\times$ the window’s traffic; section VI-F), and it coincides with Surface A wherever ADDIS’s cap binds (section VI-G). Online e-BH escapes by a different mechanism, which we do not attack.

A. Attack surface A: within-hypothesis padding

Group evidence is $\text{Ev}(G) = F(e_1, \dots, e_m)$. Under the two-point construction an ordinary flow that does not reach the k -th calibration rank contributes e-value exactly

0, so an adversary who appends r such flows to its own group makes the controller evaluate $F(e_1, \dots, e_m, 0^r)$ — “padding” throughout means appended zero-evidence flows. Nothing about the detector changes: the malicious flows still individually exceed the ceiling. The attack is aimed at the trust layer, not the classifier, and is available to anyone who can open ordinary connections. We consider three merging classes: the symmetric mean (a valid e-merging family at every arity, the subject of Theorem 10); the pre-committed cap $\sum e/n_0$ (valid only for $m \leq n_0$); and pre-committed asymmetric position-weighted rules (section VI-E).

B. Symmetric padding impossibility

Definition 9 (τ -padding-robustness, attainment). A family $\{F_N\}_{N \geq 1}$ of e-merging functions is τ -padding-robust, for $\tau > 1$, if for every $m \geq 1$, every finite $x \in [0, \infty)^m$ with $F_m(x) \geq \tau$, and every $r \geq 0$, $F_{m+r}(x, 0^r) \geq \tau$; it attains τ if some finite input has $F_m(x) \geq \tau$.

Attainment rules out a vacuous escape: without it the definition is satisfied by rules that never reach τ , such as $F \equiv 1$, which can never fire since rejection needs evidence $\geq 1/\alpha_t > 1$.

Theorem 10 (Symmetric padding impossibility). No family of symmetric e-merging functions that attains τ is τ -padding-robust, for any $\tau > 1$. Quantitatively, for every symmetric e-merging family, every finite $x \in [0, \infty)^m$ and every $\tau > 1$,

$$F_{m+r}(x, 0^r) < \tau \quad \text{whenever} \quad r > \frac{\sum_i x_i}{\tau} - m.$$

Proof sketch. Vovk and Wang’s Proposition 3.1 [11] shows the arithmetic mean essentially dominates any symmetric e-merging function: $F(e) > 1$ implies $M_K(e) \geq F(e)$ on finite inputs. Ordinary domination is false ($(\lambda + (1-\lambda))M_K$ is a counterexample), so it is used through a case split, giving $F_N(x) \leq \max(1, \text{mean}(x))$. Padding gives $\text{mean}(x, 0^r) = (\sum x)/(m+r)$, which falls below τ once $r > (\sum x)/\tau - m$; attainment supplies an x with $F_m(x) \geq \tau$. The full proof, a tightness argument, and an independent special case needing no domination result are in appendix A. $\square$

Since rejection requires evidence $\geq 1/\alpha_t$ with $\alpha_t < 1$, the threshold is always $\tau > 1$, so Theorem 10 applies to every rejection threshold any valid online procedure can set. The one symmetric rule that resists it is the pre-committed cap $\sum e/n_0$: valid and padding-robust within its cap, but not an e-merging family above n_0 — its validity is the condition $m \leq n_0$ — and below the cap strictly dominated by the mean. It escapes by being inadmissible, and the power it forfeits is a measured detection loss (section IX-D and appendix F).

And appending is not the only lever. Theorem 10 is about appending to a hypothesis, so a statistic that appending cannot lower — the group max of section IX-B — resists it outright. It does not resist the complementary

move: creating hypotheses ahead of the target, which appends nothing and pushes it out of the cold-start window past which no rejection is possible. Because group-level calibration shrinks the ceiling, that window collapses from 748–902 steps to 65–86, and every group-max detection falls to 62–295 inserted hypotheses (table XXXV). Getting ahead of a target is free under first-flow arrival; under a public hash it costs an offline search we price at 1.0×10^4 – 1.8×10^5 candidate keys, on three conditions we state rather than assume — a public seed, enough endpoint pairs the adversary can actually send between, and a stable key encoding, which our cache’s category codes are not; under a keyed hash the ranking cannot be computed offline at all, so the adversary must insert blind — and we measure what that costs (table XXXVI). The seed does not stop the attack: all 84 detections across the window-order cells that detect anything fall in every draw at a budget of 10^6 keys. What it changes is the currency. A public hash buys the ranking offline for free and instantiates only the $G^* = 30$ –813 keys that land ahead; a keyed seed forces the adversary to instantiate every trial as real traffic between distinct endpoint pairs, and at matched reliability that is 2.1–89 $\times$ more host pairs (median 19 $\times$) — N_{99} runs to 3.4×10^4 . That cost is a capability, not a computation, and the result assumes an adversary who has it: N_{99} counts endpoint pairs it must actually be able to originate traffic between, so on a network where only a few hundred reachable pairs are available to it the seed is not a partial mitigation but a hard stop — which is the regime in which a keyed hash is worth most, and which our cost figures should not be read as covering. A keyed canonicalization is therefore a genuine partial mitigation for targeted insertion, and it does nothing whatever to Surface A padding, which needs no key at all. Validity by design plus padding robustness therefore does not buy attack resistance — it buys resistance to one attack while making the other far cheaper. On the shipped pipeline the two levers split: insertion beats padding for one alert at none of the twelve window-order cells, but silences the entire controller for 217–813 hypotheses against 80 – 7.9×10^8 flows to pad every detection separately — the latter an upper bound, since suppressing an early rejection lowers R and can make the rest cheaper or moot, and heavy-tail driven at its extreme (at 0.85 ten of 72 pad costs carry almost all the sum). Per episode the targeted insertion cost is 434–5,674 hypotheses, which is why padding wins there.

C. Real-network suppression cost

On real traffic (fig. 4A), priced against the running level $1/\alpha_t$ at each detected episode’s step, suppressing a detected alert is cheap: under the canonical order the medians are 24, 6 and 116 flows at the three windows that detect anything — 0.55, 0.62 and the known-invalid 0.85, the dearest of the three being the one that carries no guarantee — over $n = 3, 11$ and 34 individually-priced

episodes, so read the first two as prices rather than as rates (the three at 0.55 cost 23, 24 and 33) — and under the first-flow upper bound — which detects more episodes, deeper into the level sequence — 118, 72, 1,851, 1,165 and 5,202, heavy-tailed to 10^7 – 10^8 for late, low- α_t alerts. The order changes which episodes survive to be attacked, not whether they can be: across 50 orders the per-order median stays at 16–178 flows across both detector seeds. Pricing against the static $\tau = T/w_0$ — the first-step level, a conservative stand-in for e-LOND’s cheaper T/α — is the cheap extreme (median 34 flows at 0.85 under the canonical order, 33 under first-flow); we report the larger running-level cost.

The realism of the padding barely matters. Four of the five pools (generic benign, attacker-origin, protocol-matched, black-box) coincide within a few percent at each window (table XIV), because the running level (median $\sim 10^4$ – 10^6) dwarfs the largest pool mean (216.8); only service-matched is 1.4–2.0 $\times$ dearer. The black-box pool uses network-observable frequency alone — the most common (proto,port) over the traffic preceding deployment, with no labels or detector output — so it is the weakest-attacker rule, yet also the correct one: taking the mode over the deployment window would instead pick the attacker’s own flood, which does not dilute (table XXI).

Two attacker capabilities must be kept apart. Constructing the pad is black-box: it needs no detector access, only an ordinary service the victim answers (0 firings in 20,000). Sizing it to the minimal $r^* = \lfloor S\alpha_t \rfloor - m + 1$ is not — that uses the realised evidence sum S , the running level, and the unperturbed trajectory — so every reported r^* is an oracle lower bound. A grey-box attacker who knows the design but not the live state must over-provision: since $1/\alpha_t$ spans $\sim 10^4$ – 10^6 within a window, a fixed budget Y suppresses exactly the alerts with $r^* \leq Y$, so guaranteeing an arbitrary one requires budgeting for the tail — equally for every pool, so pool-insensitivity stands.

Can the attacker put the pad flows where they have to go?: The episode key is (SrcIP, DstIP, bucket), so a pad must land on the target’s own host pair — the attacker sends ordinary traffic to the host it is already attacking. A label-matched pool cannot be built here (all 310 host pairs carrying any malicious flow are 100% malicious), but structure settles the endpoint question: the detector reads Protocol and 32 timing/volume statistics but no endpoint identity, so endpoint identity cannot affect the score. What it does not settle is response-dependent transfer — 17 of 33 features are backward or bidirectional — so whether an ordinary flow of a service the victim answers transfers is empirical. We test it directly: 20,000 real black-box-service flows through the shipped detector fire 0 (a one-sided 95% upper bound of 1.5×10^{-4} on the firing rate), and injecting them dilutes $E(G) = S/(m+r)$ to suppression at exactly the closed-form r^* on every detection under both orders — 3/3 and 34/34 under the canonical order the paper reports, 18/18 and 72/72 under the first-flow upper bound

(table XXI). The canonical arm is the one that carries the argument, since only there does padding provably leave every other $\mathcal{M}_j$ untouched (section IX-D); the first-flow arm shows the result is not an artefact of the smaller set.

D. Does the attack transfer? A second detector and a second dataset

The suppression above is scoped to flow-level features, so we probe that scope. First we build the host-conditioned detector the boundary calls for — six strictly-causal host-context features on a second, higher-AUROC detector (0.954 vs 0.916; table XXIII). It sharpens C1: the higher AUROC buys fewer e-LOND detections (2 against 18, both first-flow), so ranking quality and operational alerting move in opposite directions. The transfer still holds at both windows — replaying the append causally, black-box pads fire zero and the cost stays the flow-level closed form ($r^* = 108$ against 118, both under first-flow arrival, the order that experiment runs under) — but LSPR23 cannot settle the benign-inclusive question, its attacked pairs being 100% malicious.

We therefore re-measure on AIT-LDSv2.0 [22], whose eight organisations serve real benign-to-victim traffic. The full Surface A chain — leave-one-org-out flow detector, in-org benign calibration, e-LOND, and suppression by replaying real benign flows — suppresses 78 of 79 detected episodes under the canonical order (russellmitchell excepted), and 84 of 85 when the same folds are resequenced by first-flow arrival (santos excepted; table XXVI); a cross-organisation host detector adds only a weak signal (8.9% on shaw). Host conditioning does not stop the attack once the pad’s own context is modelled: recomputing the six features as pads accumulate, all nine host-conditioned detections across shaw and wilson suppress (tables XXIII to XXV; boundary limits in section X-A).

E. Can asymmetric aggregation fix it?

The natural escape is to abandon symmetry: a pre-committed, position-indexed weighting $F(e) = \sum_i w_i e_i$ is valid under arbitrary dependence whenever $\sum_i w_i \leq 1$ [23], and is padding-invariant by construction. Two exact bounds close it.

Theorem 11 (Reach). For a pre-committed weight sequence with $\sum_i w_i \leq 1$, the number P of positions at which a lone attack flow can fire the rule is at most $\alpha_t M$ — the same quantity that governs the feasibility horizon of Corollary 4.

Theorem 12 (Front-load cost). The number of leading zero-evidence flows that make detection impossible is $L^*(w) = \min\{L : \sum_{i>L} w_i < \beta\}$ with $\beta = 1/(\alpha_t M)$, which is finite for every summable w .

Theorem 12 is the same summable-tail lemma as Proposition 3, on the within-episode position axis rather than stream time — one result on two axes, which is why no

choice of weights escapes: the design sets only the price, and it is paid out of power.

Measured at the primary window (table XV), every padding-invariant scheme that retains the mean’s recall is defeated by 1 to 8 leading flows, against 111 appended for the symmetric mean on the same episodes — 14–112 $\times$ cheaper (buying more robustness, L^* up to 164, forfeits $\sim 58\%$ of the recall, first-flow on both legs); the stress window has the same shape. Two schemes fail structurally: last-event-only is not padding-invariant, and a security-prior port weighting is out of class and fires on none of the 275 episodes. So a pre-committed scheme cannot have padding invariance, usable power, and ordering robustness at once — validity forces $\sum w \leq 1$, Theorem 11 caps the reach, Theorem 12 makes L^* finite. The asymmetric escape only converts appended padding (hundreds to thousands of flows) into front-loading (one to a few hundred); it does not stop the attack.

F. Attack surface B: controller-state manipulation

Section IV-C showed that ADDIS escapes the feasibility horizon because its spending index counts hypotheses that are selected but not candidates, which threshold conformal evidence at $k = 1$ never produces here. That escape is what exposes it: an adversary who manufactures episodes with $p \in (\lambda, \tau]$ advances the index at will, and such episodes are never rejected, so they cost it no discoveries of its own.

Theorem 13 (ADDIS state budget). Under $\gamma_j = (j + 1)^{-1.6}/\zeta(1.6)$ (ADDIS’s 0-indexed sequence), k -th-rank conformal evidence, and precursors injected at index $D = 0$, B^* is the smallest D with $D + 1 > [(\tau - \lambda)W(|\mathcal{C}| + 1)/(k\zeta(1.6))]^{1/1.6}$, where $W = w_0$ before the first rejection and $W = \alpha R$ after R of them.

Verified against the implementation on the real stream at position 0.85, the known-invalid window — necessarily so, for the reason two paragraphs below (fig. 4B): $B = 202$ precursor episodes leave all 152 rejections intact, and $B = 203$ leaves none — permanently, an absorbing state of exactly the kind ADDIS was designed to avoid. The attack survives the grey-box setting: the admissible group size is a τ/λ -wide interval, so a $\pm 33\%$ estimate of $|\mathcal{C}|$ suffices and unknown controller state costs only $36\times$ over-provisioning.

Does this reach online e-BH? No: it escapes Propositions 2 and 3 by a fixed point over the whole history, not a spending index, so there is nothing for injected evidence to advance. Whether some other manipulation of its fixed point exists is open; the result here is about the index-conditioning mechanism, exemplified by ADDIS.

Is this an artefact of the invalid window? ADDIS’s horizon-escape on two-point conformal evidence is inseparable from its guarantee-invalidity: the spending index fails to advance precisely when the p-values never land in $(\lambda, \tau]$, the same reason they are not uniformly conservative — so no such stream is both escape-feasible and guarantee-valid. We therefore confirm the mechanism on

a synthetic stream with genuinely conservative $\text{Unif}(0, 1)$ nulls, where ADDIS’s guarantee holds (empirical all-null $\text{FDR} = 0.010 \leq q$; table XIX). The attack is then adversarial state poisoning — injecting admissible alternative precursors (non-null, so ADDIS’s true-null assumption is untouched and its guarantee holds on the attacked stream) — and still drives every detection to silence, at a median $B^* = 140$ over 300 guarantee-valid streams (range 140–255, the tail from a rare null false positive that transiently restores wealth). The security consequence is a property of the mechanism, not of the 0.85 window’s validity failure.

The price is set by the calibration budget, not any implementation weakness. One precursor must be a group large enough that its single firing clears $\lambda - \lfloor \lambda(|\mathcal{C}|+1)/k \rfloor + 1 = 453,279$ flows — so silencing the controller costs 9.2×10^7 flows, scaling as $\lambda|\mathcal{C}|[(\tau - \lambda)w_0|\mathcal{C}|]^{0.625}/k^{1.625}$. The same quantity that buys feasibility buys resistance to the state attack.

In operational units, the attack is conspicuous, not cheap. Those 92 million flows are 36.1 GB over position 0.85’s 2.4-hour span — 33.4 Mbit/s, four hosts at 10 Mbit/s each. What makes it structural is its volume against the monitored population: $37.5\times$ the window’s entire flow count, $5.6\times$ the whole dataset. Suppressing one e-LOND alert, by contrast, costs a median 116 flows under the canonical order — 4.7×10^{-5} of the window’s traffic, almost six orders of magnitude below, and 5,202 flows (2.1×10^{-3} , four orders) under the first-flow upper bound.

G. The two attack surfaces coincide

ADDIS caps its level at λ and the state-advancement window starts at λ , so wherever the cap binds the cheapest suppressing pad also advances the spending index (101 of ADDIS’s 147 true detections): the state attack comes free with the padding one (cap-saturation implies landing; appendix A). It is also cheaper against ADDIS, whose level is five orders of magnitude above e-LOND’s — padding one episode costs a median 3.4×10^7 flows, so state manipulation wins from the third episode on. The large per-hypothesis level that makes ADDIS escape is what makes it cheap to defeat through the state that maintains it: a design lesson, not a theorem for every procedure.

VII. Experimental Methodology

A. Dataset and protocol

LSPR23 [24], captured at Locked Shields, the largest live-fire cyber-defence exercise: 16,353,511 flows over 161.5 hours, 1,644,599 of them malicious (10.06%; 0.48–0.81% of episodes) — a live-fire base rate that section IV’s prevalence sweep prices as an optimistic bias rather than leaving as a caveat. The stream is timestamp-sorted before any split, and 89% of flows fall in the final 25.6 hours, so window positions are chronological slices rather than random splits. It carries no flow-to-campaign labels — `Expoid_dst` is empty for 1,630,732

of the 1,644,599 malicious flows, and Category, Severity and SigID are effectively constant where present, one code covering 99.98% of the labelled rows — which is why our units are episodes. A separate red-team task record (288 narratives, 83 machine-readable compromise reports, 295 timestamped step submissions) is external ground truth for the alert audit of section IX-A only, never a feature.

Chronological protocol. Training, then calibration, then deployment, no leakage. Fixed-size blocks (calibration 15%, test 15% of N) are slid to five positions along the sorted stream, so T and $|\mathcal{C}|$ vary because the traffic differs, not the window sizes. Benign subsampling is confined to training. Two detector seeds per position vary the scores while leaving T and $|\mathcal{C}|$ untouched, so the feasibility margin is seed-independent by construction and seeds are never pooled. Across positions, $|\mathcal{C}|$ runs 1,813,113–2,449,031.

Which windows carry the headline numbers, and why that is not a validity claim. 0.55 and 0.62 were pre-committed as the primary and replication windows before any benign-tail diagnostic existed; no post-hoc rule selects them, and we do not invent one. The diagnostic runs in the other direction, and it is weak by construction: the entire split rests on 0–46 benign firings out of 1.6–2.3M (1, 1, 3, 2, 46 at seed 0; zero at two windows on seed 1), so at 0.55 the exact Clopper–Pearson interval on the measured/nominal ratio is $[0.03, 5.96]$ — an interval that excludes essentially nothing. Four positions are not refuted (1.07 – $3.79\times$, every interval containing 1; table XXX), which is a failure to reject and not support. Position 0.85 alone is a clear violation ($50.9\times$, interval $[37.3, 67.9]$), so it is the known-invalid stress-test window, used only for mechanism and attack-cost measurement. Accordingly every guarantee-bearing statement in this paper, at every window, is conditional on Assumption 1 — the window names mark which stream carries which headline number, not which stream has been empirically validated.

B. Detectors and procedures

Detector novelty is not a contribution: the detector is HistGradientBoosting, and every detection, attack and coverage number is its output. An unsupervised Isolation Forest runs alongside as a feasibility-vs-sufficiency probe — same feasibility margin (a function of $|\mathcal{C}|, k, T$ alone), yet tail reach 0.000 at all ten configurations: not one attack flow outscores all 1.8–2.4M benign calibration scores. Feasibility is necessary, not sufficient; since it never detects, the attack results are statements about the HistGradientBoosting pipeline.

Procedures. LOND, LORD++, e-LOND, SAFFRON, ADDIS, online e-BH, and the e-GAI family (e-LORD, e-SAFFRON, mem-e-LORD), transcribed from primary sources and unit-tested against literal $O(T^2)$ implementations and four identities the papers state. Throughout $\alpha = q = 0.05$, $w_0 = 0.025$, $k = 1$, ADDIS $\lambda = 0.25$, $\tau = 0.5$; each γ -procedure runs two spending sequences: $\gamma_j \propto j^{-1.6}$ (no horizon) and horizon-uniform $\gamma_j = 1/T$

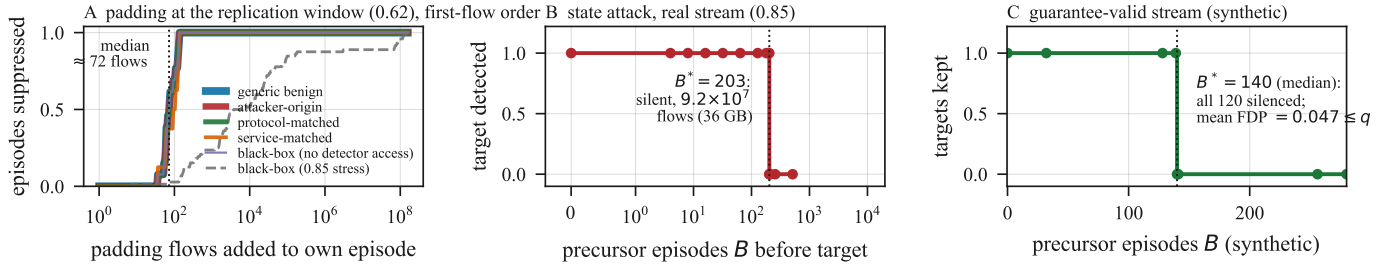


Fig. 4. What does each attack cost? A: suppressing one alert at the replication window (0.62) is a low-footprint evasion, insensitive to padding realism; the costs plotted are the first-flow arrival arm — the optimistic upper bound over the orders we audit, and the more expensive one for the attacker — over that order’s 13 detections. Under the canonical order table I reports, the same window detects 11 episodes at a median of 6 flows each (table XIV). B: permanently silencing ADDIS through its adaptive state (0.85 stress) is conspicuous in volume, at exactly the Theorem 13 threshold. C: the same poisoning on a synthetic stream where ADDIS’s guarantee genuinely holds still silences every detection.

(max-min optimal, needs the stream length, labelled [oracle]).

C. Grouping and reproducibility

The online stream. A group’s evidence uses every flow in its time bucket, so its hypothesis is emitted only when the bucket closes, using only observations available then; no group is decided using a flow outside its own closed bucket. Groups closing in the same bucket are sequenced by a pre-committed, evidence-independent key, and that tie-break is a choice with consequences (section IX-D). We report the canonical choice — a splitmix64 hash of the group’s own (SrcIP, DstIP, bucket) key, collisions broken on the key — which reads no arrival time, so padding an episode moves no other group’s slot and Assumption 1 is provably undisturbed by it. The initial first-flow arrival order is reported alongside as an optimistic upper bound over the orders we audit.

The headline grouping is (SrcIP, DstIP, 2-hour bucket) with the uncapped arithmetic-mean rule; the capped $\sum e/n_0$ rule enters only in the cap-policy analysis. This is a deployable configuration, not an oracle optimum, and is not tuned on the reported window: it is in fact near-worst among the feasible groupings under previous-window selection, the cap being the selection-stable choice while the grouping does not transfer (section IX-D), and the feasibility gate precedes selection, being a function of $|C|$, k and T alone. Every result is computed on the full stream; each closed form is verified numerically against an independent implementation, and each measurement script carries unit tests.

VIII. Operational Evaluation

A. The point the controller chooses, and its cost

Comparing a $q = 0.05$ procedure against an arbitrary classifier threshold measures nothing, so we place every method on the achievable frontier of the episode-max-score threshold family at its own budget — an ex-post oracle (its thresholds use the test labels) that bounds what the ranking admits (tables XXVII and XXVIII), under the

canonical order the headline uses. The frontier is itself order-free — it is a threshold family on the episode max score — which we verify rather than assume, since its tie-break is by stream position; only the controller rows move. The result is a clean negative: every method sits within 0.047 recall of the best achievable point at its budget, the informative comparisons being the FDP > 0 rows.

The comparison that matters is thus choice of point, not efficiency, and the canonical order sharpens it. At the primary window e-LOND operates at 3 alerts, recall 0.011, FDP 0.000 (18 and 0.065 under first-flow arrival), while an ex-post oracle at the same zero error reaches recall 0.378 on 104 alerts — thirty-five times as much, against six under the optimistic order: a measure of how far below the ranking’s ceiling the α -wealth process sits, and one the ordering objection makes larger rather than smaller. And the lever that moves the point is not operational: sweeping q twenty-fold moves median recall by 0.088, while switching the spending sequence at fixed q moves it by 0.188 ($5.8\times$ the alert count) — a first-flow paired sweep, both legs on the same stream. Formal error control can therefore reduce the operator’s freedom over the workload–recall tradeoff. Closing the loop does not rescue this unless labels arrive in minutes: the three feedback threshold controllers (P, PI, AQT; appendix B, first-flow) reach FDP 0.402 (P, PI) and 0.523 (AQT) at fifteen-minute disposition delay, agree with one another by four hours, and reach the no-feedback threshold (0.841) only by eight (table XXIX) — a statement about delay on these controllers, not about feedback-driven testing in general, for which folding revealed labels into the threshold with finite-sample control retained [25] is the principled construction whose input our measurement prices.

IX. Robustness and Failure Analyses

A. Calibration validity and the labels

Calibration drift is mild almost everywhere. The measured-to-nominal firing ratio stays near one across our ten position $\times$ seed configurations (median 1.94 $\times$; independently 0.86–2.03 $\times$ on the AIT wilson testbed [22]), with

a single exception — the stress-test window 0.85, where the benign firing rate runs $50.9\times$ nominal, so $\mathbb{E}[e] \approx 51$ and the evidence is not a valid e-value there. The excess is sharply localised: three host pairs carry 44 of the 46 firing benign flows, and removing them restores the ratio to $2.22\times$. It is best explained by label error, which the alternatives — distribution shift, a new benign mode, duplicated records, Mondrian conformal — each fail to account for (appendix H), though we do not prove it.

The marginal rate is the weakest possible evidence for Assumption 1, so we interrogate it conditionally. We stratify the firing benign flows on the observable components of $\mathcal{M}_j$ — arity (what padding moves), service, bucket, and destination host — and read them at rank depths $k \in \{1, \dots, 1000\}$, resampling both the shared calibration tail and the test side (table XXXI). The result cuts both ways. At the depth the pipeline uses ($k = 1$) the data mostly cannot speak: at the shipped depth 129 of 134 strata carry no test at all (428 of 536 cells across the four depths), and none that are estimable reject at a primary window — an inconclusive non-rejection, not support. Deeper in the tail it is another matter: 32 contrasts survive correction at all five windows (25 after removing the one confounder we can identify), the strongest an arity bin at 0.77 — uncomfortable, since arity is what padding moves. But a departure at $k = 1000$ is compatible with a valid rank-1 rate, and no finite sample identifies a conditional expectation, so we draw the conservative conclusion: the guarantee-bearing claims stay stated under Assumption 1, not supported by this diagnostic.

Two consequences follow. First, the guarantee does not attach at 0.85, so that window serves only for mechanism and attack-cost measurement; the feasibility horizon (eq. (3)) is unaffected, not depending on the evidence at all. Second, every FDP there is measured against labels: auditing the ADDIS ($\gamma \propto j^{-1.6}$) set’s 152 alerts against the red team’s task record, 146 agree (96.1%) and six disagree, so an under-estimate is not excluded. e-LOND’s realised FDP is 0.0066 (frozen) or $[0, 0.0265]$ (strict post-hoc); the audit is partly circular (99 of 152 verdicts rest on the label-generating identities), so it indicates the error’s direction, not a value. Two of the audit’s four indicators are not discriminative — temporal concurrency separates 100.0% of alerts from 86.0% of non-alerted benign episodes — so only the strict rule’s two branches carry weight.

What Assumption 1 does and does not govern. It is worth separating this once, because the three results of this paper do not stand or fall together. (1) The feasibility theorems need no distributional premise at all. Propositions 2 and 3 and Corollaries 4 and 5 are statements about a level sequence against a bounded evidence ceiling; they hold whatever the evidence distribution is, valid or not, and eq. (2) is a function of $|C|$, k and T alone. C1 is therefore untouched by any reading of this diagnostic. (2) Theorem 10 is an aggregation theorem, not a data assumption. It is a statement about symmetric e-

merging functions — that no such function which can fire is invariant to appended zero-evidence arguments — and it is proved from admissibility, with no reference to how the evidence was generated. C2’s impossibility half is likewise independent. (3) Only the FDR numbers on the empirical stream are conditional on Assumption 1: the realised FDPs, the guarantee-bearing reading of any of them, and Lemma 1’s marginal validity. Those are the claims that carry the premise, and the ones we mark accordingly throughout. And the construction that would discharge (3) by design is priced in (1): calibrating the grouped unit directly replaces Assumption 1 with ordinary group exchangeability and then, by Corollary 5, cannot supply the calibration it needs — an infeasibility that holds however the premise is resolved (section IX-B).

B. Calibrating the group directly

The diagnostic interrogates Assumption 1; the constructive alternative is to remove the need for it, by running split conformal on the grouped unit — same key, same protocol, on the calibration split — so that only calibration/test group exchangeability is required. Built and measured over two group statistics, five windows and two seeds, it fails all three counts a repair must meet (table XXXII).

The repair really does discharge Assumption 1: the premise becomes ordinary calibration/test group exchangeability, and the group statistic’s dependence on arity does not by itself contradict that — a statistic may be strongly arity-dependent while calibration and null test groups remain perfectly exchangeable. Two things qualify it, and neither is a refutation. Group exchangeability is still a deployment assumption, and our threat model is one under which the test groups are partly adversary-constituted: insertion instantiates endpoint pairs that enter the stream as true nulls, so the group-formation process itself is something the adversary moves. (Padding does not bear on it: it enlarges the attacked group, which is a false null, and the premise constrains only true nulls.) That is a reason the assumption may be fragile under group-formation shift, not a proof that it fails. And the max statistic makes that fragility acute — its rank correlation with arity is $+0.23$ – $+0.41$ across the ten window-seed cells, and within the calibration set groups above a thousand flows contribute 3.6–15.3% of their own bin to the top 1% against 0.26–0.44% for singletons. That is a decomposition of the calibration set, and marginally 1% by construction, so it settles nothing on its own; scoring benign test groups against the calibration threshold does. Marginal coverage is roughly right (0.55 – $2.32\times$ nominal over the ten cells), while arity-conditional coverage is not: strata run from $0.13\times$ to **$24.2\times$** nominal under max against 0.00 – $5.6\times$ under the mean. The guarantee is marginal over an arity mix, and arity is the stratum the adversary chooses. Stratifying calibration on arity restores a per-stratum guarantee and leaves each stratum a cold-

start window of 2–52 hypotheses. The robustness and the fragility are one fact: appending cannot lower a maximum, which is why Theorem 10 does not run against it and why the arity dependence is there at all — while the mean, arity-invariant to within noise (-0.12 to $+0.09$), is exactly the rule Theorem 10 defeats.

Grant the repair its validity premise, then: it still cannot alert. Group calibration replaces 1.8–2.4M flows with 37–57k groups, so M falls 40–49 $\times$ at unchanged T : the margin moves $+0.067 \rightarrow -0.977$, no window is feasible, and detection at the primary and replication windows falls from 18 and 13 to zero. That is the boundary, not a weak statistic — at 0.55 the cold-start window is 81 hypotheses while the first of 110 firing groups arrives at position 401, and feasible-step counts match $\lfloor (\alpha M(R+1)/\zeta(1.6))^{1/1.6} \rfloor$ everywhere, 0.85 included, where 42 detections are the bootstrap (10.5 $\times$) on evidence that is not a valid e -value anyway. By Corollary 5 that is 20 groups per hypothesis the construction cannot supply. Two scope notes: calibration groups containing any malicious flow are excluded, which drops up to 26.1% of calibration flows, in groups whose mean arity runs up to 45.4 $\times$ the benign mean — a bias that favours firing, so the negative result survives its own bias; and the worst arity bin reaches 15.3 $\times$ nominal.

C. Calibration contamination

At $k = 1$ the conformal rule is one order statistic, the calibration maximum, so contamination acts only through it. Injecting j attack flows above that maximum hands the threshold to the attacker once $j \geq k$, so tolerable adversarial mislabels are exactly $k - 1$, and feasibility has fixed $k = 1$. One mislabelled flow above the calibration maximum takes episode recall to 0.000, from 0.011 at 0.55 and 0.133 at 0.85 under the canonical order (0.065 and 0.282 under first-flow arrival; the mechanism is order-free, the recall it destroys is not). A random mislabel is milder but not much: median recall 0.004 at 0.55 and 0.035 at 0.85, with zero recall in 22.5% and 32.25% of draws, of scale $\varepsilon^* = 1/(Nq_0) = 1.11$ –1.31 flows. Contamination tolerance is a count, not a rate, and adversarially zero.

A second channel runs the other way, and gently: a contaminating flow below the calibration maximum only inflates the p -value denominator, degrading the guarantee to at most $q(1 + \varepsilon)$ — doubling q would take a contaminating set the size of the calibration set. So power dies at one flow while validity barely moves. And the feasibility margin is not a safety indicator: across the sweep that destroys detection it does not move at all ($+0.436$ throughout, while recall falls $0.133 \rightarrow 0.000$ canonically, $0.282 \rightarrow 0.000$ under first-flow); it rises to $+0.556$ only in a separate below-threshold injection that leaves every detection standing.

D. Order robustness and configuration selection

Ordering is robust to timestamp ties (at most 1.3% of episodes share a first timestamp; all 36 reported statistics are invariant across 50 randomised within-timestamp orders).

The within-bucket order, by contrast, is load-bearing — and it is where Proposition 2 becomes visible. Since α_t decays with step index and evidence is capped at $(|C|+1)/k$, only a prefix can reject at all: 902/902/865/824/748 steps at the five windows, 1.6–2.6% of the stream. A detection therefore needs a near-ceiling episode to land inside it — which under first-flow arrival is exactly what live-fire attack traffic does. Sequencing the same episodes by their own metadata instead takes the counts from 18/13/30/31/72 to 3/11/0/0/34, and at 0.70 and 0.77 no episode clears its own step anywhere in the stream. The ordering result and the feasibility theorem are one finding measured twice: three detections out of 275 malicious episodes at the primary window is Corollary 4’s 20 $\times$ shortfall seen from the detection side. Over 50 pre-committed orders the count has median 1 here and 23.5 at the stress window; the canonical order is not a uniformly typical draw from that ensemble — above its median at three windows, at its maximum at 0.62, below it at the two where nothing detects — while first-flow matches or exceeds its maximum at every window and seed — strictly, except where nothing detects at all (table X).

Why that canonical order. First-flow arrival is an attacker timing lever — delaying an episode’s first flow pushes it to a later, lower- α_t slot — and the damage escapes the attacker’s own hypothesis: one pad at the start of its bucket moves 373 other episodes here, shifting the γ_j (hence the $\mathcal{M}_j$) each is offered. A tie-break reading only the group’s own (SrcIP, DstIP, bucket) key makes that 0 by construction, so section II’s Assumption 1-preservation argument now describes the order the paper reports, not one it merely recommends. Three qualifications, all measured rather than assumed away. It removes the timing lever, not the metadata one: the sort key is own-key-only, so appending flows moves no other group, but a group’s absolute rank still depends on which other keys exist, and an adversary that creates keys moves ranks. A public hash is moreover grindable over the attacker-selectable (SrcIP, DstIP) — with the feasible prefix at 1.6–2.6%, an adversary trying 100 candidate keys lands one inside it with probability 0.80–0.93, and avoids it on the first draw with probability ≥ 0.974 ; the canonical order is therefore a canonicalization, not a security mechanism. A keyed hash removes grinding, at the price of a seed the adversary can neither learn nor infer — one our published seed does not instantiate. It is a real but partial defence: blind insertion still suppresses every detection, at 2.1–89 $\times$ the instantiated-key cost (section VI and table XXXVI). And it is not free of power: one keyed instance detects 0 here.

The frozen configuration is not tuned on its reported window: under previous-window selection and five leave-one-out folds it is near-worst of the 30 groupings feasible at every window (of 35 in the grid), and the cap transfers across windows while the grouping does not (worst normalised regret 0.070 vs 0.991). The cap is deliberately the validity-and-cost choice, not the powerful one.

E. What did not work

Several natural alternatives fail: inflation and fragmentation (per-group evidence is flat in group size), boosting ($b^* = 1$ for two-point evidence), raising k , smoothing, asymmetric weighting, stratified (Mondrian) conformal, and the published α -death fix — each for the reason given in appendix I.

X. Discussion

What should a practitioner do? If deployment-wide FDR matters, treat Corollary 4 as a sizing requirement: the calibration corpus is linear in the hypothesis count, unattainable at flow granularity, so choose the alerting unit first and the procedure second. If fine resolution matters, restarting restores detection but promises only a per-epoch guarantee (or deployment-wide at less power) — the SOC should be told which. If grouping is used, treat the aggregation rule as adversarially exposed, and fix the within-bucket order by group metadata rather than arrival. If analyst labels arrive quickly (minutes, not hours), feedback control is reasonable, but the construction to reach for folds labels into the threshold with error control retained [25], not a bare threshold controller. And because these observations are attacker-generated whenever the grouping key is attacker-choosable, any adaptive mechanism that conditions future behaviour on observed data — aggregation and controller state alike — must be threat-modelled.

Scope. We do not claim online FDR control is unusable, that dependence breaks it (e-values remove that), or that generic drift invalidates conformal evidence; nor that a better detector cannot help — detector quality cannot change C1’s margin, though it can change power. Four things we specifically do not claim: that the conformal-resolution incompatibility was itself unknown (C1 characterises when it becomes structural [13]); that Surface B reaches online e-BH; that host-aware detection is no defence against padding in general; or that Assumption 1 is established rather than assumed.

Four scope statements attach to specific results and travel with them. (i) Proposition 6 bounds donation’s level before the first rejection only; past $1/\delta$ rejections the level is not bounded by that expression at all, and the bound must not be quoted as a global cap. Runs here do pass that count; what holds instead is the weaker and sufficient fact that the denominator never vanishes on these streams (section IV-C). (ii) For e-TOAD’s decision-deadline construction we use the interpretation consistent

with the stated limiting cases ($d_t = t \Rightarrow$ e-LOND, $d_t = \infty \Rightarrow$ online e-BH); the alternate literal interpretation is included as a sensitivity analysis (table XIII) and does not affect our conclusion. (iii) One equation of the compound-e construction admits two readings — the rejection history it scores is not nested, so a literal snapshot and a running union differ. We report both and default to the literal one; the choice does not affect our conclusion. (iv) The canonical within-bucket order is a canonicalization, not a security mechanism: the insertion attack survives both a public hash, which is grindable, and a keyed one, which is not — under a secret seed the attack costs $2.1\text{--}89\times$ more instantiated endpoint pairs but still succeeds. That measurement assumes the target sits in the first bucket of its window, which holds at every cell here; where an earlier bucket exists the adversary inserts there and precedes the target with probability one.

A. Limitations

A single live-fire dataset. Five window positions are five quasi-independent deployments of one exercise, not five datasets; a triage of ≈ 40 public datasets against two catalogues found none with a genuine flow-to-campaign identifier, complete negatives and recoverable order, so our units are episodes. Position 0.85 is not a valid-e-value window: every FDP there is measured against labels, and the audit indicates but does not prove an over-estimate. Assumption 1 is assumed, not established: the metadata-stratified diagnostic fails to reject it at the shipped depth — where 129 of 134 strata cannot be tested at all — but finds the conditional statement measurably imperfect deeper in the tail, at every window (table XXXI). It is nonetheless a priced choice rather than a convenience: the construction that would discharge it silences the primary and replication windows outright, whatever one grants about the premise it substitutes (section IX-B). We do not claim to have exhausted the alternatives — a two-way calibration split isolating granularity from the group statistic, and grouping keys the adversary cannot choose, are both untested. Per-alert padding costs are oracle lower bounds, and the other attack quantities are labelled where they appear rather than covered by that phrase (section II); the state attack’s absolute cost is dataset-specific in $|C|$ and window span, its closed form is not.

Padding placement is scoped to flow-level features (section VI-D): the full Surface A chain suppresses 78 of 79 real detected episodes across the eight AIT organisations under the canonical order (84 of 85 under first-flow), and on the nine host-conditioned detections we replay end-to-end, conditioning does not stop it once the pad’s own context is modelled — but we do not claim host-aware detectors are no defence in general, since the accumulation model pins distinct-peer counts (padding from new hosts is uncovered), the six features are one design, and the high-volume flood regime is untested.

Scope of C1: discrete evidence, an uninterrupted controller, two covered families.

XI. Related Work

Online FDR and α -investing. LORD, LOND, SAFRON, ADDIS, e-LOND and online e-BH establish stream validity [3], [4], [5], [6], [7], [8], [12], [18], building on α -investing [15], safe testing [26] and offline FDR [27]. “ α -death” [5] is held cured by α -investing; it is not cured here (the published fix [16] equals e-LORD before the first rejection). Recent procedures directly applicable to our setting — strict improvements over e-LOND and r-LOND under arbitrary dependence, built from an online e-closure principle and from compound e-values [20], [21] — are classified against C1 in section IV-C rather than merely cited: neither rescues a controller that fails to obtain its first rejection before the bounded-evidence cold-start horizon, which is what makes C1 a statement about the evidence rather than about a particular controller’s vintage.

Batching. A SOC receives a batch when a time bucket closes, so a batched procedure is the natural architecture, and batching is known to recover power and stabilise ordering [28]. We do not use BatchBH or its relatives because their guarantees need independence or positive dependence within the batch, which co-located flows sharing hosts and one calibration tail cannot supply — the reason the pipeline is built on e-values at all. The batched alternative that survives arbitrary dependence and applies directly here is a bucket-close decision deadline, which we run (section IV-C and table XIII); and batching with a deployment-wide budget obeys Corollary 4 unchanged, since allocating $\sum_b \alpha_b \leq \alpha$ across batches reproduces the same exchange rate. Batching escapes the horizon only by resetting the guarantee per batch, which is our restart escape under another name (section IV-D).

Conformal resolution meets online testing: the nearest prior work. Conformal prediction [9] and conformal p-values for outlier testing [10] supply the evidence, with a calibration-conditional adjustment for small- k failure. Resolution collapse under discrete conformal evidence was studied in a batch setting with smoothing as the remedy [14], and batching framed as power recovery [28]; our restart analysis instead prices the guarantee’s composition. The collision our C1 must be positioned against is Huo et al. [13], who state that conformal p-values are bounded below by $1/(|\mathcal{C}| + 1)$, which “leads to unsatisfactory performance for online multiple testing methods based on p-values”, and who observe α -death empirically for LOND, SAFRON and ADDIS on conformal evidence, easing as $|\mathcal{C}|$ grows, responding with a local-FDR selection rule rather than a compositional analysis. The phenomenon is thus known; what C1 adds is when it stops being a power loss and becomes structural — the finite absorbing horizon, exchange rate, escape classification and escape costs of section I, for two families covering the e-value procedures

as well as the p-value ones. Krönert et al. [17] come closest in form, requiring $|\mathcal{C}| = \nu m/\alpha - 1$ for a windowed BH — our Corollary 4 algebra — but to make that windowed procedure’s realised level exact, tuning $|\mathcal{C}|$ upward for power; ours is the threshold below which no rejection is possible at all, over an uninterrupted horizon rather than a sliding window (their windowing is our restart escape, section IV-D).

Trustworthy ML intrusion detection, and online conformal-FDR pipelines. TESSERACT [29] sets the temporal-evaluation constraints our protocol respects, and Transcend(ent) [30], [31] apply conformal calibration with rolling recalibration. Closest to ours are pipelines that compose conformal evidence with sequential false-discovery control [32], [33], [34], [35], with Gao et al. [36] at the decision boundary but in a batch regime. These establish that such pipelines can be made valid; our subject is a different failure mode. Label quality is a known problem [37], [38], so we audit against an external record.

Online testing with analyst feedback. Our feedback baselines are threshold controllers, not FDR procedures; the principled comparison is the dedicated literature that folds revealed labels into the testing threshold with finite-sample control retained — GAIF/OCTF [25] under instant, delayed, full and bandit feedback, and online conformal selection under irreversible decisions [39]. Those are orthogonal to what we measure: our variable is the wall-clock disposition delay, and our numbers price how quickly a SOC must adjudicate for any feedback signal to be worth having.

Adversarial testing and aggregation. E-merging theory gives the admissible symmetric [11] and asymmetric [23] rules, both fixing the arity padding violates. Adversarial testing games [40], batch FDR under score perturbation [41], conformal novelty detection under evasion [42] and Byzantine nodes reporting corrupted p-values [43] all reach the pipeline through the evidence; our adversary perturbs no score and corrupts no report, manipulating a hypothesis’s composition and the controller’s state with traffic alone.

XII. Conclusion

Finite conformal evidence and uninterrupted control create a finite discovery horizon, and the calibration a deployment would need to stay inside it exceeds what these streams supply by orders of magnitude. Coarsening the alerting unit buys feasibility on an exact exchange rate, paying in resolution; restarting buys it back at a price the design chooses — the deployment-wide guarantee, or some power. The aggregation this requires is adversarially exposed, and a procedure escaping the horizon on observed evidence exposes a second, priced surface. Our own FDR statements are no exception to the argument: each is conditional on Assumption 1, and the window whose evidence is not a valid e-value carries no guarantee at

all. These guarantees must be evaluated as properties of the composed system.

Open Science

We will release the complete set of artifacts needed to reproduce every number and figure in this paper, and they exist and have been run in full at the time of submission: a single executable notebook that carries the paper’s derivations and measurements alongside the prose, the library of measurement scripts it imports, and the cached results of a full end-to-end run. The notebook has two modes. In its default mode it renders every table and figure from shipped results in seconds, so a reviewer can read the implementation next to the claim it produces without obtaining any data. In full mode it recomputes every number from the raw flow data, and we have executed it in that mode end to end to confirm that it reproduces the shipped results.

The obstacle is third-party data, of which the artifact now depends on two records. LSPR23 is not ours to redistribute but is publicly available under CC-BY-4.0 (Zenodo record 8042347), and the artifact ships the exact preprocessing commands that turn the published CSV into the pipeline’s inputs. The host-conditioned transfer test (table XXIV) additionally uses AIT-LDSv2.0 [22] (Zenodo record 5789064, CC-BY-4.0): the artifact ships the per-organisation tstat netflow extraction and the exact malicious/benign TCP label mapping (appendix C) that turn its released packet captures into the eight-organisation netflow inputs, following the dataset’s own processing recipe. Full-mode reproduction therefore requires downloading both records; the derivations, procedure implementations, attack code, and cached results carry no such restriction.

They will be deposited in a fully anonymised repository within three days of submission, and on Zenodo for the camera-ready.

LLM Usage Considerations

Disclosure and editorial use. LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality.

Responsibility. The execution of this project entailed dozens of distinct experiments and attack measurements, a volume that exceeded what the authors could manually develop within the project’s timeframe. Consequently, Large Language Models (LLMs) were utilized to assist in generating the majority of the automation and evaluation scripts, under strict, detailed prompting and oversight by the authors. It is important to emphasize that the core methodology proposed in this paper requires no LLM training or operational LLM usage. The total expenditure for LLM API token usage during development was \$347.04.

Accountability and correctness. The authors have done extensive testing while developing the scripts and verified

every number in this paper, and are responsible for any mistakes there may be.

Transparency. The released artifact contains the same code the results were computed with, so any claim in the paper can be checked against its implementation directly.

Ethical Considerations

This paper describes working attacks against a statistical trust layer for intrusion detection, so we state our reasoning against the Menlo Report framing [44].

Respect for persons and justice. All measurements are made on public research datasets released for this purpose, captured during authorised cyber-defence exercises on purpose-built ranges rather than on production networks carrying third-party traffic. We rely on the publishers’ own ethical review and release terms and did not conduct a separate consent audit, so we make no claim about individual consent beyond what those releases state. No attack in this paper was executed against any live network, and no personal data is used, disclosed, or re-identified.

Beneficence. The attacks target a class of system that is, to our knowledge, not yet widely deployed, which is the point at which disclosure is cheapest and most useful. We disclose no new weakness in any detector: both attacks follow from properties the relevant procedures publish about themselves — the symmetry of an e-merging rule, and the index that drives a spending sequence — and an adversary reading those papers would reach the same conclusions. We report the attacks’ costs precisely, including the ones that make them expensive, so that defenders can reason about which of them are worth mitigating rather than being left with a qualitative warning.

Law and public interest. We identify a small set of host pairs whose labels we believe to be wrong, and deliberately do not publish corrected labels: rewriting a public dataset’s labels from our own detector’s tail would be circular, and the pairs were identified using test-split information. We report the direction of the error instead.

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Appendix A Proofs

A. Group-level marginal e-validity (Lemma 1)

Proof. For a tested group G_j , recall $\mathcal{M}_j$ is the σ -field generated by the pre-committed metadata that determines this hypothesis: the grouping key, the membership indicators $\mathbf{1}\{i \in G_j\}$, the arity m_j , the group’s position in the pre-committed stream order (and hence the spending weight γ_j it is offered), and the horizon T . It carries no information about any other group’s composition or arity, and — as the scope note below records — none about the realised calibration scores. Fix a true-null group G_j (all constituent flows benign) with $m_j > 0$. By linearity

of conditional expectation and $\mathcal{M}_j$ -measurability of the membership indicators and of m_j ,

$$\begin{aligned}\mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j] &= \mathbb{E}\left[\frac{1}{m_j} \sum_{i \in G_j} e_i \mid \mathcal{M}_j\right] \\ &= \frac{1}{m_j} \sum_{i \in G_j} \mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1,\end{aligned}$$

the inequality by Assumption 1. Taking expectations and applying the tower property gives $\mathbb{E}[\text{Ev}(G_j)] = \mathbb{E}[\mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j]] \leq 1$. No independence or exchangeability among the $\{e_i\}$ is used, so within-group dependence is unrestricted; and because e-LOND and online e-BH control FDR under arbitrary dependence given marginally valid per-hypothesis e-values, arbitrary between-group dependence is permitted as well. An empty group ($m_j = 0$) tests no hypothesis; assigning it the trivial $\text{Ev} = 1$ is harmless. For the two-point conformal e-value $e_i = M \mathbf{1}\{K_i \leq k\}$ the premise $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ is exactly $\Pr(K_i \leq k \mid \mathcal{M}_j) \leq k/(|\mathcal{C}|+1)$, since $M = (|\mathcal{C}|+1)/k$. $\square$

Why the conditioning is per hypothesis. A single global metadata σ -field $\mathcal{M}$ will not support the attack analysis of section VI. $\mathbb{E}[e_i \mid \mathcal{M}] \leq 1$ does not imply $\mathbb{E}[e_i \mid \mathcal{M}'] \leq 1$ for a richer $\mathcal{M}'$ — take $e = 2\mathbf{1}_A$ with $\Pr(A) = \frac{1}{2}$: against the trivial σ -field the conditional expectation is 1, against $\sigma(A)$ it is 2 on A — so an adversary who enriches the stream’s metadata anywhere would, on the global reading, void the premise even for groups it never touched. With $\mathcal{M}_j$ the statement is local and survives: padding adds flows carrying the attacked group’s own key, so it changes no other group’s key, membership or arity, and adds no hypothesis, leaving T fixed. Provided the within-bucket order is a function of group metadata alone (the deterministic hashed-key canonical order of section IX-D, with collisions broken on the key itself), every true-null group’s position and γ_j are likewise fixed, so $\mathcal{M}_j$ is literally the same σ -field before and after the attack. Under an arrival-ordered tie-break this fails: a pad placed before the attacked group’s first flow moves that group within its bucket and can shift a true-null group’s index, hence γ_j , hence $\mathcal{M}_j$. Creating a new host pair is likewise outside the padding claim — it adds a hypothesis and changes T , which the horizon-uniform $\gamma_j = 1/T$ reads.

Scope. The lemma certifies the e-value controllers (e-LOND, online e-BH) only. The p-value procedures carry their own conditions — conditional super-uniformity for mFDR, and independence with monotonicity for FDR — which this argument does not supply, and which section IV-C shows ADDIS violates on this stream. The premise $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ is marginal over the calibration draw and test score; it is not validity conditional on the realised calibration scores, which are deliberately excluded from $\mathcal{M}_j$ (the Bates adjustment supplies that stronger, calibration-conditional guarantee at a power cost and is not required here). $\mathcal{M}_j$ is also smaller than a global metadata σ -field, so Assumption 1 is the weaker of the two conditional premises, while remaining stronger than unconditional $\mathbb{E}[e_i] \leq 1$.

B. Feasibility (Propositions 2 and 3, Corollary 4)

Proposition 2. With $\alpha_t = \alpha \gamma_t g(\mathcal{H}_{t-1})$, $g \leq c(R_{t-1} + 1)^d$, and $R_{t-1} \leq t - 1$, we have $\alpha_t \leq \alpha c_g \gamma_t t^d$. A rejection at t needs $\alpha_t \geq 1/M$, hence $\gamma_t t^d \geq 1/(\alpha c_g M)$. If $\gamma_t t^d \rightarrow 0$ then only finitely many t satisfy that inequality, since a sequence tending to zero exceeds a fixed positive constant finitely often. A finite set of indices has a maximum, so there is a last feasible index t^* (or none at all), and every $t > t^*$ fails the necessary condition: the run is absorbed from $t^* + 1$. No monotonicity of γ is needed. Requiring permanence from the first infeasible step would be a stronger property — false for non-monotone γ , since feasibility can fail once and return — and it is not what section IV defines. $\square$

Proposition 3. Let $\alpha_t = w_0 \gamma_t + \sum_j a_j \gamma_{t-\tau_j}$ with $0 \leq a_j \leq \alpha$ over rejection times $\tau_j < t$. During a rejection-free run of length Δ the last rejection is at least Δ before t , so every lag $t - \tau_j \geq \Delta$ and the lags are distinct. With $\Gamma(\Delta) = \sum_{u \geq \Delta} \gamma_u$ we have $\gamma_{t-\tau_j} \leq \Gamma(\Delta)$ termwise and $\gamma_t \leq \Gamma(t) \leq \bar{\Gamma}(\Delta)$, so, using $a_j \leq \alpha$ and the distinctness of the lags, $\alpha_t \leq w_0 \Gamma(\Delta) + \alpha \Gamma(\Delta) = (w_0 + \alpha) \Gamma(\Delta)$. Since Γ is non-increasing and $\sum \gamma < \infty$ forces $\Gamma(\Delta) \rightarrow 0$, the quantity $\Delta^* = \min\{\Delta : (w_0 + \alpha) \Gamma(\Delta) < 1/M\}$ is finite; Δ only grows during a rejection-free run, so the state is absorbing. No monotonicity of γ is used. $\square$

Corollary 4 and max-min optimality of $\gamma_t = 1/T$.

Before any rejection, family I has $\alpha_t = \alpha \gamma_t$ and family II has $\alpha_t = w_0 \gamma_t$; write the cold-start coefficient $c_0 \in \{w_0, \alpha\}$, so $\alpha_T = c_0 \gamma_T$. Feasibility at T requires $\alpha_T \geq 1/M = k/(|\mathcal{C}|+1)$, i.e. $|\mathcal{C}| \geq k/\alpha_T - 1 = k/(c_0 \gamma_T) - 1$. Under $\gamma_T = 1/T$ this is $|\mathcal{C}| \geq kT/c_0 - 1$: $kT/w_0 - 1$ for a level- w_0 procedure such as LORD++, and $kT/\alpha - 1$ for LOND/e-LOND. For optimality, a cold-start rejection is possible at every $t \leq T$ iff $\min_{t \leq T} \gamma_t \geq k/((|\mathcal{C}|+1)c_0)$; among sequences with $\sum_{t \leq T} \gamma_t \leq 1$ the value $\min_{t \leq T} \gamma_t$ is maximised, at $1/T$, by the uniform allocation, since any other allocation places some $\gamma_t < 1/T$. Hence $\gamma_t = 1/T$ maximises the feasible horizon. $\square$

C. The UAI 2026 procedures (Propositions 6 and 7)

Proposition 6. Donation e-LOND sets $\alpha_t = \delta \gamma_t (|R_{t-1}| + 1)/(1 - (\delta(|R_{t-1}| + 1)\bar{W}_t \wedge 1))$ with $\bar{W}_t = \sum_{i \in R_{t-1}} \gamma_i ((E_i - \frac{1}{\delta \gamma_i (|R_{t-1}| + 1)}) \wedge 1) + \sum_{i \notin R_{t-1}} \gamma_i (E_i \wedge 1)$. with both sums running over $i < t$. Every summand is γ_i times a quantity capped at 1, so $\bar{W}_t \leq \sum_{i < t} \gamma_i \leq 1$ always. On a rejection-free prefix $R_{t-1} = \emptyset$, so for $\delta < 1$ the denominator is at least $1 - \delta > 0$ and $\alpha_t \leq \delta \gamma_t / (1 - \delta)$, which is Proposition 2 with $c_g = \delta/[\alpha(1 - \delta)]$ and $d = 1$: Proposition 2 writes the level as $\alpha \gamma_t g(\mathcal{H}_{t-1})$, so the constant must absorb the ratio δ/α , and it reduces to $1/(1 - \delta)$ exactly in the $\delta = \alpha$ setting we run.

Why $\delta < 1$ is required. At $\delta = 1$ the denominator is $1 - \bar{W}_t$, which the donated wealth drives to zero, and the conclusion fails outright. Take $\delta = 1$, $\gamma_t = 2^{-t}$ and $E_t = 1$

for every t . Then $\bar{W}_t = \sum_{i < t} \gamma_i (E_i \wedge 1) = \sum_{i < t} \gamma_i = 1 - 2^{-(t-1)}$ and $\alpha_t = \gamma_t / (1 - \bar{W}_t) = 2^{-t} / 2^{-(t-1)} = \frac{1}{2}$ for every t . The run is rejection-free because rejection needs $E_t \geq 1/\alpha_t = 2$ and $E_t = 1$ — this is why the evidence is pinned at exactly 1 rather than merely bounded below by it — yet the offered level never decays, so with any $M \geq 2$ every step stays evidence-feasible and the run is never absorbed. The proposition therefore quantifies over $\delta \in (0, 1)$; our runs use $\delta = \alpha = 0.05$, where $1/(1 - \delta) = 1.05$.

The restriction to the rejection-free prefix is not cosmetic. With $R = |R_{t-1}|$ the denominator is $1 - (\delta(R + 1)\bar{W}_t \wedge 1)$, which vanishes once $\delta(R + 1)\bar{W}_t \geq 1$; since $\bar{W}_t$ can approach 1, that is reachable at $R + 1 \geq 1/\delta$. The bound is therefore a cold-start statement, which is exactly the state Corollary 4 concerns. It is also sharp: taking $E_i \geq 1$ for every i makes $\bar{W}_t = \sum \gamma_i \rightarrow 1$ and the bound tight. $\square$

Proposition 7. Closed e-LOND sets $\alpha_t = \min_{S \subseteq [t-1]: D_t(S) > 0} \delta \gamma_{|S|+1} (|R_{t-1}| + 1) / D_t(S)$ with $D_t(S) = 1 + |S \cap R_{t-1}| - \delta E_S (|R_{t-1}| + 1)$, where E_S aggregates the evidence on S . Because α_t is a minimum over admissible subsets, any single admissible S furnishes an upper bound.

Take $S = \{i < t : E_i = 0\}$, of size Z_t . A hypothesis carrying zero evidence is never rejected, so $S \cap R_{t-1} = \emptyset$; and $E_S = 0$ because the aggregator is zero on all-zero input. Hence $D_t(S) = 1 > 0$, S is admissible, and $\alpha_t \leq \delta \gamma_{Z_t+1} (|R_{t-1}| + 1)$.

This is the decay of Proposition 2 with the time index rescaled from t to $Z_t + 1$. The bound itself is unconditional and pathwise; the conclusion that the state is absorbing is not, and Corollary 8 supplies its hypothesis. $\square$

Corollary 8. By Proposition 7, $\alpha_t \leq \delta \gamma_{Z_t+1} (|R_{t-1}| + 1)$ along any run. On a rejection-free run $|R_{t-1}| = 0$, so $\alpha_t \leq \delta \gamma_{Z_t+1}$. If $\gamma_{Z_t+1} \rightarrow 0$ then $\alpha_t \rightarrow 0$, and since the evidence is bounded above by M (section IV) the offered level falls below $1/M$ after finitely many steps and no rejection is possible thereafter: the state is absorbing, which is Proposition 2's conclusion.

Form (i), exact and finite-horizon. Along the rejection-free run $|R_{t-1}| = 0$, so $\alpha_t \leq \delta \gamma_{Z_t+1}$. Our $\gamma_j \propto j^{-1.6}$ is decreasing, so no rejection is possible once $\gamma_{Z_t+1} < 1/(\delta M)$, i.e. once $Z_t \geq H - 1$ for $H = \min\{j : \gamma_j < 1/(\delta M)\}$, which is exactly Proposition 2's index horizon. Write $P_T = \#\{i \leq T : E_i > 0\}$. Of the $t - 1$ hypotheses before step t , at most P_T carry positive evidence, so $Z_t \geq t - 1 - P_T$ for every $t \leq T$ — a counting statement, with no assumption on the stream. Hence $Z_t \geq H - 1$ as soon as $t \geq H + P_T$, and

$$t_{\text{closed}} := \inf\{t : Z_t \geq H - 1\} \leq H + P_T.$$

The closure therefore reaches the absorbing state within an additive P_T steps of the horizon Proposition 2 gives, whenever $H + P_T \leq T$. Nothing is asymptotic and no

density is assumed. For a non-monotone γ the same conclusion follows without the decreasing step, because α_t is a minimum over admissible subsets: any $H - 1$ zero-evidence indices furnish an admissible S and hence the same bound.

Form (ii), asymptotic. Suppose $\liminf_t Z_t/t = \rho > 0$ and fix any $\rho' < \rho$. By the definition of the $\liminf$ there is a t_1 with $Z_t \geq \rho' t$ for all $t \geq t_1$; so $Z_t \rightarrow \infty$ and, $\gamma_j \propto j^{-1.6}$ being decreasing to 0, $\gamma_{Z_t+1} \leq \gamma_{\lceil \rho' t \rceil} \rightarrow 0$. Substituting into Corollary 4's discovery-horizon computation replaces the index t by $\rho' t$: the effective spending index is rescaled by ρ' , so absorption is reached once both $t \geq t_1$ and $\rho' t \geq H - 1$, i.e. by $\max\{t_1, \lceil (H - 1)/\rho' \rceil\}$. The onset t_1 cannot be dropped: a positive $\liminf$ constrains only the eventual density and permits an arbitrarily long initial transient. The time to absorption is therefore stretched by $1/\rho'$, not shortened — a sparser zero-evidence stream buys the closure more steps, not fewer. This holds for every $\rho' < \rho$ and not, from this argument, at ρ itself: the $\liminf$ gives eventual domination below its value, not at it.

On our streams form (i) is the operative one and needs no density at all. The number of hypotheses carrying any evidence is $P_T \in \{0, \dots, 152\}$ over the ten window-seed cells, at most $\hat{p} = 4.82 \times 10^{-3}$ of T , attained at position 0.85, and $P_T = 0$ at 0.70 seed 1; so the horizon offset is at most 152 hypotheses and the cap is attained (ratio 1.000000 over every configuration measured). The argument is specific to sparse evidence: a stream in which most hypotheses carry positive evidence has P_T close to T , Z_t small, and this bound says little — which is the regime the closure was designed for, and not a security stream. $\square$

D. Symmetric padding impossibility (Theorem 10)

We give a full proof (Route A) and an independent special-case argument (Route B) that needs no domination result; Route B establishes the impossibility for a negatively-dependent construction, not the arbitrary- x quantitative bound, which is Route A's.

Route A (essential domination). Vovk and Wang's Proposition 3.1 [11] states that the arithmetic mean M_K essentially dominates every symmetric e-merging function: $F(e) > 1 \Rightarrow M_K(e) \geq F(e)$. Ordinary domination is false ($\kappa + (1 - \kappa)M_K$, for any $\kappa \in (0, 1)$, is a symmetric valid counterexample), so we use it through a case split: for finite x , either $F_N(x) \leq 1$, or $F_N(x) > 1$ and then $M_K(x) \geq F_N(x)$; in both cases $F_N(x) \leq \max(1, \text{mean}(x))$. Padding $x \in [0, \infty)^m$ with r zeros gives $\text{mean}(x, 0^r) = (\sum_i x_i)/(m + r)$, so

$$F_{m+r}(x, 0^r) \leq \max\left(1, \frac{\sum_i x_i}{m+r}\right) < \theta \text{ once } r > \frac{\sum_i x_i}{\theta} - m.$$

Attainment supplies an x with $F_m(x) \geq \theta$ to which this applies, so no attaining symmetric family is θ -padding-robust for any $\theta > 1$. $\square$

Route B (a negatively-dependent special case). Fix $\theta > 1$ and m . Draw a uniformly random size- m subset $S \subseteq [N]$ and set $e_i = (N/m)\mathbf{1}\{i \in S\}$. Then $\mathbb{E}[e_i] = (N/m)\Pr(i \in S) = 1$, so each e_i is an e-variable, and the vector is a permutation of $((N/m)\mathbf{1}_m, 0^{N-m})$. A symmetric F_N maps every realisation to the same constant, and a constant e-variable is ≤ 1 . Hence $F_N((N/m)\mathbf{1}_m, 0^{N-m}) \leq 1 < \theta$ for all $N \geq m$: appending $N - m$ zeros to the m firing positions drives any symmetric rule below θ . $\square$

Tightness and attainment. The bound is tight: the mean itself has $F_{m+r}(x, 0^r) = (\sum x)/(m+r) \geq \theta$ exactly while $r \leq (\sum x)/\theta - m$. Attainment (section VI) is necessary, not cosmetic: the constant rule $F \equiv 1$ is symmetric, valid and padding-invariant but never reaches any $\theta > 1$, satisfying θ -padding-robustness vacuously; since rejection needs evidence $\geq 1/\alpha_t > 1$, restricting to attaining families excludes exactly the rules that cannot fire.

E. Reach and front-load (Theorems 11 and 12)

Let $w = (w_1, w_2, \dots)$ be precommitted with $w_i \geq 0$ and $\sum_i w_i \leq 1$, let $F(e) = \sum_i w_i e_i$ with each $e_i \in \{0, M\}$, and write $\beta = 1/(\alpha_t M)$ for the weight mass a detection needs: a lone flow firing at position i gives $F = w_i M$, which clears $1/\alpha_t$ iff $w_i \geq \beta$.

Theorem 11. The positions at which a lone attack flow fires are $\{i : w_i \geq \beta\}$. Since $\sum_i w_i \leq 1$, at most $1/\beta$ indices can carry mass $\geq \beta$, so $P = |\{i : w_i \geq \beta\}| \leq 1/\beta = \alpha_t M$, the quantity that governs Corollary 4. $\square$

Theorem 12. Prepending L zero-evidence flows moves every attack flow to a position $> L$. Every $e_i \leq M$, so $F(e) \leq M \sum_{i>L} w_i$, and detection requires $\sum_{i>L} w_i \geq \beta$; it is impossible once $L \geq L^*(w) = \min\{L : \sum_{i>L} w_i < \beta\}$. Summability gives $\sum_{i>L} w_i \rightarrow 0$, so the set is non-empty and L^* is finite for every summable w .

Sufficient, and exact only under a full-firing episode. The displayed bound is attained only when every position $> L$ fires, so L^* is the exact minimum in that case and an upper bound otherwise. The gap is real: for $w = (0.4, 0.3, 0.3, 0, \dots)$ and $\beta = 0.5$ the tail first falls below β at $L = 2$, yet an episode carrying a lone firing flow is already undetectable at $L = 0$, since $\max_i w_i = 0.4 < \beta$ and Theorem 11 gives $P = 0$. We therefore quote L^* as an attacker's upper bound throughout. $\square$

Consequently a precommitted scheme cannot simultaneously be padding-invariant (weights fixed by absolute position), have usable power (some $w_i \geq \beta$), and make ordering robustness unbounded ($L^* = \infty$). Concentrating weight early gives $L^* = 1$ but $P = 1$; spreading it over more than $1/\beta$ positions drives every $w_i < \beta$, so $P = 0$.

F. ADDIS state budget (Theorem 13) and the coincidence

Theorem 13. ADDIS offers $\alpha_t = \min\{\lambda, (\tau - \lambda)[W\gamma_{S^t - C_{0+}} + \dots]\}$. The theorem assumes $\lambda \geq k/(|\mathcal{C}|+1)$, i.e. the cap sits at or above the conformal floor, so that

the driving term and not the cap is what the attack has to push below $1/M$; if the cap were already below the floor no target could be rejected at $D = 0$ and the budget would be zero rather than the displayed value. With $W = w_0$ before the first rejection and $W = \alpha R$ after R of them, and 0-indexed $\gamma_j = (j+1)^{-1.6}/\zeta(1.6)$. Precursors injected at index 0 hold the spending index at $S^t - C_{0+} = D$, so the driving level is $(\tau - \lambda)W\gamma_D$. The collapse of ADDIS's reward sum to the single term $W\gamma_D$ is exact in the pre-rejection state, where $W = w_0$ and there is one active term; after R rejections each reward term carries its own selected-non-candidate lag, and $W = \alpha R$ bounds their sum, so what follows is then an upper bound on B^* rather than its value. The attack is mounted in the pre-rejection state, since the precursors precede every target. A rejection is possible only while $(\tau - \lambda)W\gamma_D \geq 1/M$, i.e. $(D+1)^{-1.6}/\zeta(1.6) \geq k/((\tau - \lambda)W(|\mathcal{C}|+1))$, i.e. $D+1 \leq [(\tau - \lambda)W(|\mathcal{C}|+1)/(k\zeta(1.6))]^{1/1.6}$. The smallest D violating this is B^* . Each precursor is a group carrying exactly one firing flow, so its evidence is M/m and its p-value m/M ; landing in $(\lambda, \tau]$ needs $m \in (\lambda M, \tau M]$, whose smallest integer is $m^* = \lfloor \lambda(|\mathcal{C}|+1)/k \rfloor + 1$ (the interval is non-empty because $(\tau - \lambda)M \gg 1$). The total is B^*m^* , scaling as $\lambda|\mathcal{C}|[(\tau - \lambda)w_0|\mathcal{C}|]^{0.625}/k^{1.625}$. $\square$

Coincidence. When ADDIS's cap binds, $\alpha_t = \lambda$, and the minimal suppressing pad raises the padded p-value just past λ . Because the p-value moves in steps of one conformal floor $1/(|\mathcal{C}|+1) \ll \tau - \lambda$, it lands in $(\lambda, \tau]$ and advances the index: cap-saturation implies landing. The converse is not forced, but on this stream every unsaturated detection has a level four orders of magnitude below λ , so none can land, and the observed equality (all 101 saturated detections land) is the generic case rather than a coincidence of this window.

G. Restart FDR composition

Proof. Restart runs n controllers on disjoint epochs, each controlling its own FDR at q . The deployment-wide proportion FDP = $\sum_i V_i / (\sum_i R_i \vee 1)$ is a weighted average of the per-epoch FDP _{i} with data-dependent weights $R_i / \sum_j R_j$, so per-epoch control does not bound $\mathbb{E}[\text{FDP}]$. Indeed, suppose each epoch makes one false rejection with probability q and no rejection otherwise. Each epoch then has $\text{FDR}_i = \mathbb{E}[V_i / (R_i \vee 1)] = q$, yet whenever any epoch rejects every pooled rejection is false, so the deployment-wide FDP is 1 on that event and 0 otherwise, and with independent epochs $\text{FDR} = \Pr(\text{some epoch rejects}) = 1 - (1-q)^n$. (The neighbouring construction — each epoch always rejecting once, false with probability q — instead gives pooled $\text{FDR} = q$, so non-composition requires the all-or-nothing epoch above and does not follow from per-epoch control alone.) In contrast, mFDR pools: if $\mathbb{E}[V_i] \leq q\mathbb{E}[R_i]$ for every i , the median inequality gives $\sum_i \mathbb{E}[V_i] / \sum_i \mathbb{E}[R_i] \leq q$.

This non-composition is specific to budget reset — every epoch spends its own q , total nq . A preallocated

TABLE II

Terms held distinct throughout the paper. Definitions only, so order-invariant: no quantity here is measured on a sequenced stream. Validity is an assumption; feasibility, power, and resolution are separate consequences.

Term	Meaning
Statistical validity	the assumptions the stated FDR/e-value guarantee needs (Assumption 1 and Lemma 1)
Feasibility	whether any admissible evidence can cross the controller threshold
Detection power	whether malicious episodes actually generate sufficient evidence
Episode recall	recall under the current grouping definition (moving denominator)
Atomic coverage	coverage of a fixed fine-grained attack-unit denominator
Alert blur	number of fixed atomic reference units one issued alert represents
Padding cost	flows needed to suppress a current alert (Surface A)
State-attack cost	flows/precursors to manipulate the sequential controller state (Surface B)

restart — pre-committed epoch budgets $q_1, \dots, q_n$ with $\sum_i q_i \leq q$, fixed before deployment — instead preserves the deployment-wide guarantee. Since $\sum_j R_j \geq R_i$, the pooled proportion is dominated termwise,

$$\text{FDP} = \frac{\sum_i V_i}{\sum_j R_j \vee 1} \leq \sum_i \frac{V_i}{R_i \vee 1} = \sum_i \text{FDP}_i$$

(both sides 0 when no epoch rejects), so by linearity of expectation $\text{FDR} \leq \sum_i \text{FDR}_i \leq \sum_i q_i \leq q$ — no independence across epochs is required (independence enters only the $1 - (1 - q)^n$ counterexample above). Each e-LOND epoch delivers $\text{FDR}_i \leq q_i$ under the same group e-validity premise (Assumption 1); the shared calibration set only induces between-epoch dependence, which this bound tolerates, provided the epoch boundaries and the q_i are not data-dependent. Restart therefore does not simply void the deployment-wide guarantee: it forces a choice — budget reset recovers more power but controls only a per-epoch FDR, whereas preallocation keeps deployment-wide $\text{FDR} \leq q$ at a power cost (table XII). $\square$

Appendix B

Procedure Definitions

Each procedure offers a level α_t and rejects H_t when the evidence exceeds $1/\alpha_t$ (p-value procedures reject when $p_t \leq \alpha_t$). We write R_{t-1} for the rejection count, τ_j for the j -th rejection time, and, for the candidate-based procedures, $C_{j+}(t)$ for the number of candidates after τ_j . Transcriptions were unit-tested against literal $O(T^2)$ implementations of the published formulas and against four identities the sources state (offline e-BH as the $t = K$ case of online e-BH; $R(\text{e-LOND}) \subseteq R(\text{online e-BH})$; e-LORD = e-LOND under $\gamma_t = \omega_t \prod_{j < t} (1 - \omega_j)$; e-SAFFRON($\lambda=0$)=e-LORD).

LOND, e-LOND: $\alpha_t = \alpha \gamma_t (R_{t-1} + 1)$,

LORD++: $\alpha_t = \gamma_t w_0 + (\alpha - w_0) \gamma_{t-\tau_1} + \alpha \sum_{j \geq 2} \gamma_{t-\tau_j}$,

SAFFRON: $\alpha_t = \min \{ \lambda, (1 - \lambda)[w_0 \gamma_{t-C_{0+}} + \dots] \}$,

ADDIS: $\alpha_t = \min \{ \lambda, (\tau - \lambda)[w_0 \gamma_{S^t - C_{0+}} + \dots] \}$,

online e-BH: reject H_i iff $E_i \geq 1/(k_i^* \alpha \gamma_i)$,

e-LORD (e-GAI): $\alpha_t = \omega_t (\alpha - \sum_{j < t} \frac{a_j}{R_{j-1} + 1}) (R_{t-1} + 1)$.

SAFFRON's candidates are the $p \leq \lambda$; ADDIS additionally selects the $p \leq \tau$ and counts selected non-candidates. Here $k_i^* = \max \{ k \leq t : \sum_{j \leq t} \mathbf{1}\{E_j \geq 1/(k \alpha \gamma_j)\} \geq k \}$ is the online-e-BH step-up fixed point, and ω_t is the e-GAI

weight update [16]. Throughout $\alpha = q = 0.05$, $w_0 = 0.025$, $k = 1$; ADDIS uses $\lambda = 0.25, \tau = 0.5$, SAFFRON $\lambda = 0.5$, e-SAFFRON $\lambda = 0.1$, and e-GAI $\phi = \psi = 0.5$, $\omega_1 = 1/T$. The two spending sequences are $\gamma_j \propto j^{-1.6}$ (normalised by $\zeta(1.6) = 2.2858$, no horizon needed) and horizon-uniform $\gamma_j = 1/T$ (max-min optimal, [oracle]).

The three feedback baselines of table XXIX are threshold controllers, not FDR procedures. Each carries a single state $u_t \in [0, 1]$, a position in the calibration score distribution: it alerts on episode t when that episode's maximum flow score exceeds the u_t -quantile of the calibration scores, and moves u_t only when an analyst's disposition of an earlier alert arrives, after a wall-clock delay L . Writing $\widehat{\text{FDP}}_t$ for the false fraction over the last 200 dispositions received so far and $\varepsilon_t = \widehat{\text{FDP}}_t - q$, the updates are

$$\text{P: } u_{t+1} = u_t + \eta_P \varepsilon_t,$$

$$\text{PI: } u_{t+1} = u_t + \eta_P \varepsilon_t + \eta_I \sum_{r \leq t} \varepsilon_r,$$

$$\text{AQT: } u_{t+1} = u_t + \eta[(1-q)\mathbf{1}\{\text{false}\} - q\mathbf{1}\{\text{true}\}],$$

with the PI integral term clamped against windup and u clipped to $[0, 1]$ throughout. P is proportional control on the estimated FDP error, PI adds the integral term, and AQT (adaptive quantile tracking) is the label-driven stochastic-approximation step whose fixed point is the u at which a q fraction of alerts is false — so AQT needs no window, updating on each disposition individually. Gains are chosen per window, delay and controller by grid search ($\eta_P \in \{0.005, 0.02, 0.05, 0.2\}$, window $\in \{50, 200\}$, $u_0 \in \{0.99, 0.999\}$, AQT $\eta \in \{10^{-4}, 10^{-3}, 5 \times 10^{-3}, 0.02\}$, $\eta_I = \eta_P/10$) minimising $(|\text{FDP} - q|, -\text{recall})$ on the realised test labels. That selection is an ex-post oracle, so the reported controller performance is an upper bound on what a deployment could achieve — which is the conservative direction for the negative conclusion drawn from it. None of the three carries a finite-sample error guarantee: they are the operational baselines a SOC would actually build, and the experiment asks what disposition delay does to them, not how they compare with the feedback-aware FDR procedures [25] that do carry one.

Appendix C

Dataset and Preprocessing

LSPR23 [24] (Zenodo 8042347, CC-BY-4.0) is a single CSV of 16,353,511 flows. The detector reads Protocol plus 32 per-flow timing and volume statistics (columns 9–40); the grouping key reads SrcIP, DstIP (columns 2–3) and, for the service and subnet families, the ports (columns 4–5); the alert audit of section IX-A reads the annotation columns (host/segment/service labels). Timestamps are in microseconds and 7,381,261 rows arrive out of order, so the pipeline sorts on load. Splits are chronological: with test-window start fraction pos, $i_2 = \lfloor \text{pos} \cdot N \rfloor$, $i_1 = i_2 - \lfloor 0.15N \rfloor$ and $i_3 = \min(N, i_2 + \lfloor 0.15N \rfloor)$ give training $[0, i_1)$, calibration $[i_1, i_2)$ and deployment $[i_2, i_3)$; benign subsampling is confined to $[0, i_1)$. The exact awk recipes and a deterministic cache build are in the released artifact.

AIT-LDSv2.0. The transfer tests of section VI-D use AIT-LDSv2.0 [22] (Zenodo 5789064, CC-BY-4.0), eight organisation testbeds whose released packet captures we convert to per-organisation tstat tcp_complete net-flows following the dataset’s own recipe. Label mapping. Each flow carries the dataset’s own attack-label string; we treat as malicious exactly the attack-step labels it defines (service_scan, host_discover_dmz/_local, online_cracking, wpscan, dirb_scan, upload_rce_shell ...) by exact string match — the labels are atomic, so no substring matching is used — and as benign exactly the labels the dataset marks benign; flows whose label is unknown or empty are dropped rather than assumed benign, since an unlabelled flow is not evidence of innocence. Numeric columns are kept per fold only if they are non-null on $> 90\%$ of rows and non-constant; all identity fields and all absolute-time fields are excluded so no fold can key on the capture epoch, and the six host-context features are the same strictly-causal ones as on LSPR23. Splits are chronological within each organisation: calibration uses benign flows strictly preceding that organisation’s first attack flow, deployment is everything from that instant on.

Appendix D Full Experiment Matrices

Table IV gives the ten-configuration detection matrix, table V the power of every procedure over the same grid, table VI the grouping families at the primary window, table VII the joint (q, γ) sweep, table VIII the cross-window transfer of the grouping and cap, table IX the grouping-independent resolution cost against the fixed atomic evaluation reference, and table X the within-bucket order-sensitivity of the headline detection count.

Appendix E Boundary Experiments

Table XI gives the full smoothing arm (all merges) and table XII the restart arm (all epochs and allocations). Table XIII places the newest strict improvements over e-LOND, and the decision-deadline family joining our two escape routes, against the horizon.

Appendix F Attack Experiments

Table XVI gives the per-window suppression cost (running-level vs static), table XVII the aggregation-cap sweep, table XVIII the ADDIS state-attack budget, table XIX the state attack on a synthetic guarantee-valid stream, table XX both attacks in operational units, table XXI the controlled padding-dilution demonstration on the shipped detector, table XXIII the host-conditioned detector and the padding-transfer boundary, and table XXV the full Surface A suppression pipeline on the AIT benign-inclusive testbed.

Appendix G Operational Baselines

Table XXVII gives the matched operating points at the primary analysis window and table XXVIII at the stress-test window, and table XXIX the three feedback controllers under wall-clock disposition delay.

Appendix H Data-Quality Analyses

Table XXX gives the per-window benign firing rates, table XXXI the metadata-stratified reading of them that interrogates Assumption 1 conditionally, table XXXVII the adjudicated alert audit, and table XXXVIII the Bates calibration-conditional adjustment.

Appendix I Negative Results

Inflation and fragmentation fail analytically. Group evidence is the mean of per-flow e-values, so for a purely benign group j , conditioning on its metadata σ -field $\mathcal{M}_j$ (Lemma 1), $\mathbb{E}[\text{Ev}(G) \mid \mathcal{M}_j] = \frac{1}{m} \sum_i \mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1 < 1/\alpha_t$: no padding turns a benign group into a discovery (inflation). For a pure-attack group whose flows fire independently with probability ϕ , $\mathbb{E}[\text{Ev}(G)] = \phi M$ is independent of the group size m , so splitting a campaign across groups does not lower per-group evidence (fragmentation). Both are confirmed numerically.

Boosting recovers nothing for two-point evidence: the optimal e-to-e boosting factor for a $\{0, M\}$ -valued e-value is $b^* = 1$ by construction (against $\sqrt{2\tau}$ for a genuine continuous e-value), and below 1 on the realised scores. Raising k removes the operating point rather than trading power for validity; randomised smoothing removes the floor but not the reliably-detected set (section IV-D); asymmetric weighting is not a free mitigation (section VI-E); stratified (Mondrian) conformal does not repair the position-0.85 anomaly (section IX-A); and the published α -death fix (mem-e-LORD) is identical to e-LORD before the first rejection.

TABLE III

Procedures, assumptions, and coverage by Propositions 2 and 3. Power is reported at the primary analysis window (0.55, $|\mathcal{C}| = 2,448,993$, $T = 57,368$ episodes, 275 malicious) and the stress-test window (0.85, $|\mathcal{C}| = 1,813,113$, $T = 31,568$, 255 malicious), two-hour grouping.

Horizon-uniform γ requires the stream length in advance and is marked [oracle] wherever it appears. The FDP column is the 0.85 stress-window value, measured against labels where the evidence is not a valid e-value (section IX-A); at 0.55 every FDP here is ≤ 0.010 . Which procedures are covered by C1 is position-independent; the rejection counts differ because the two windows carry different traffic.

[†]SAFFRON sits outside the letter of Propositions 2 and 3 because its index counts tested hypotheses, but it does not escape in practice:

it is silent for 99.3% of the stream at 0.55 and 75% at 0.85. Only ADDIS and online e-BH are genuine escapes, at both windows — escapes of the absorbing property, which is what Propositions 2 and 3 assert: online e-BH is still unable to reject 90.1% of hypotheses at their own arrival here, exactly as often as e-LOND, and recovers them only by revisiting (section IV-C and table XIII). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

Procedure	Evidence	Spending index	Horizon needed	Covered by C1?	Rej. 0.55 (primary)	Rej. 0.85 (stress)	FDP (0.85)	Nominal guarantee / required assumptions
LOND [3]	p-value	elapsed time	no	yes (I)	18	72	0.000	FDR (super-unif. null p ; indep.)
e-LOND [8]	e-value	elapsed time	no	yes (I)	18	72	0.000	FDR under group e-validity
LORD++ [5]	p-value	lag over rejections	no	yes (II)	19	72	0.000	FDR (conditional / indep.)
e-LORD (=e-SAFFRON $_{\lambda=0}$) [16]	e-value	elapsed time (reparam.)	[oracle]	yes (I)	101	150	0.020	FDR under group e-validity
mem-e-LORD [16]	e-value	elapsed time (reparam.)	[oracle]	yes (I)	87	142	0.000	controls mem-FDR
SAFFRON [6]	p-value	non-candidates	no	no [†]	0	70	0.000	assumes independence
ADDIS [7]	p-value	selected non-cand.	no	no	110	152	0.033	null-cons. violated
online e-BH [8]	e-value	fixed point over history	no	no	18	72	0.000	FDR under group e-validity

TABLE IV

e-LOND ($\gamma \propto j^{-1.6}$) across five window positions and two detector seeds, two-hour host-pair grouping. The margin is seed-independent by construction; $|\mathcal{C}|$ is the calibration size $|\mathcal{C}|$. “silent” is the fraction of steps at which the controller is structurally infeasible — the offered level α_t has fallen below $1/M$, so no rejection is possible even at the maximal admissible evidence M (it is not the fraction of steps without a rejection). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

Position	Seed	AUROC	T	$ \mathcal{C} $	margin	rejections	recall	silent
0.55	0	0.916	57,368	2,448,993	+0.067	18	0.065	90.1%
0.55	1	0.927	57,368	2,448,993	+0.067	18	0.065	90.1%
0.62	0	0.799	49,267	2,449,031	+0.243	13	0.033	90.5%
0.62	1	0.871	49,267	2,449,031	+0.243	12	0.030	90.9%
0.70	0	0.914	37,231	2,287,988	+0.536	30	0.105	80.1%
0.70	1	0.847	37,231	2,287,988	+0.536	0	0.000	97.7%
0.77	0	0.828	31,672	2,116,418	+0.671	31	0.126	77.3%
0.77	1	0.833	31,672	2,116,418	+0.671	31	0.126	77.3%
0.85	0	0.999	31,568	1,813,113	+0.436	72	0.282	65.4%
0.85	1	1.000	31,568	1,813,113	+0.436	71	0.278	65.7%

TABLE V

Procedure power over five positions $\times$ two seeds (min / median / max), two-hour grouping. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else. The uniform rows are the horizon-uniform γ , an [oracle] that must know T in advance (table II).

Procedure	γ	rejections			recall		
		min	med	max	min	med	max
LOND	$\propto j^{-1.6}$	0	24	72	0.000	0.085	0.282
LOND	uniform	0	100	151	0.000	0.273	0.576
e-LOND	$\propto j^{-1.6}$	0	24	72	0.000	0.085	0.282
e-LOND	uniform	0	100	151	0.000	0.273	0.576
LORD++	$\propto j^{-1.6}$	0	25	72	0.000	0.089	0.282
LORD++	uniform	0	100	151	0.000	0.268	0.576
SAFFRON	$\propto j^{-1.6}$	0	21.5	70	0.000	0.069	0.275
SAFFRON	uniform	0	0	0	0.000	0.000	0.000
ADDIS	$\propto j^{-1.6}$	0	106.5	152	0.000	0.274	0.576
ADDIS	uniform	0	0	0	0.000	0.000	0.000
online e-BH	$\propto j^{-1.6}$	0	31	72	0.000	0.094	0.282
online e-BH	uniform	0	105	152	0.000	0.273	0.576

TABLE VI

Grouping families at the primary window (0.55, seed 0); flow coverage is window-specific. The margin column is the level- w_0 feasibility margin (eq. (2)); the recall and coverage columns are e-LOND under the horizon-uniform [oracle] γ , whose cold-start coefficient is $\alpha = 2w_0$. $\dagger$ marks a negative level- w_0 margin; e-LOND can still fire there when that margin exceeds $-\frac{1}{2}$ (its own $\alpha = 2w_0$ cold-start bound), which is why some daggered rows carry non-zero recall/coverage — a row is 0 only when e-LOND is itself infeasible. Feasibility and detection are thus not the same condition (cf. table I). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X).

Family	bucket	margin	episode recall	flow coverage
src-dst	300 s (5 m)	-0.637 $\dagger$	0.000	0.000
src-dst	7200 s (2 h)	+0.067	0.378	0.276
src-dst	none	+0.548	0.253	0.032
src	300 s (5 m)	+1.546	0.501	0.408
src	7200 s (2 h)	+20.505	0.382	0.353
src	none	+54.964	0.286	0.350
dst	300 s (5 m)	+0.085	0.532	0.689
dst	7200 s (2 h)	+1.736	0.304	0.033
dst	none	+3.646	0.184	0.034
subnet24	300 s (5 m)	-0.451 $\dagger$	0.448	0.697
subnet24	7200 s (2 h)	+0.501	0.285	0.031
subnet24	none	+1.251	0.193	0.031
src-dport	300 s (5 m)	-0.327 $\dagger$	0.324	0.368
src-dport	7200 s (2 h)	+3.025	0.066	0.351
src-dport	none	+6.189	0.032	0.340

TABLE VII

Error target q vs the spending sequence (median episode recall).

Feasibility scales linearly in q : the measured (margin + 1)/base ratio tracks the prediction $q/0.05$. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else. The uniform column are the horizon-uniform γ , an [oracle] that must know T in advance (table II).

q	median recall		(margin + 1) ratio	
	poly	uniform	measured	predicted
0.01	0.049	0.000	0.20	0.20
0.05	0.085	0.273	1.00	1.00
0.10	0.100	0.274	2.00	2.00
0.20	0.137	0.274	4.00	4.00

TABLE VIII

Cross-window configuration transfer: the cap transfers, the grouping does not. Each cell is the previous-window normalised regret $\rho \in [0, 1]$ (0 = oracle, 1 = worst feasible config) on the evaluation window, seed 0; “—” is a flat selection window where transfer is undefined. The grouping axis is not flat — its oracle flow coverage varies by 0.676 across windows (cap: 0.253) — yet the frozen grouping still fails to transfer. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X).

Transfer fold (select $\rightarrow$ eval)	normalised regret ρ	
	cap axis	grouping axis
0.55 $\rightarrow$ 0.62	0.000	0.945
0.62 $\rightarrow$ 0.70	0.048	0.991
0.70 $\rightarrow$ 0.77	0.070	0.000
0.77 $\rightarrow$ 0.85	0.009	0.921
worst (previous-window sel.)	0.070	0.991
worst (leave-one-out)	0.048	0.945

TABLE IX

Grouping-independent resolution cost (src-dst family, e-LOND $\gamma \propto j^{-1.6}$, seed 0). The atomic evaluation reference is the set of 5-minute src-dst malicious episodes — a fixed fine-grained denominator we choose, not externally validated incident truth, since LSPR23 carries no campaign identifier — 1899 at the primary window, 1973 at the stress-test window — and its size is fixed across every alerting bucket. “cov (fixed)” is the fraction of those atomic units covered by ≥ 1 issued alert; “blur” is the mean number of distinct atomic malicious units an issued alert lumps together (the resolution lost); “recall (moving)” is the ordinary episode recall, whose denominator changes with the bucket. As the unit coarsens the fixed coverage rises while the blur climbs from 1 to ~ 20 –40: that climb, not the moving recall, is the denominator-invariant resolution cost. $\dagger$ marks a negative level- w_0 feasibility margin; e-LOND can still fire there when that margin exceeds $-\frac{1}{2}$ (its own $\alpha = 2w_0$ cold-start coefficient), which is why some daggered rows carry non-zero coverage/recall — a daggered row reads 0 only when e-LOND is itself infeasible (cf. tables I and VI). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X).

bucket	cov (fixed)	blur	recall (moving)	flow cov.
primary analysis window (0.55)				
5 m $\dagger$	0.020	1.0	0.020	0.001
30 m $\dagger$	0.065	3.6	0.077	0.004
1 h $\dagger$	0.068	7.2	0.058	0.004
2 h	0.152	16.1	0.065	0.007
6 h	0.240	25.3	0.103	0.011
1 d	0.385	38.5	0.123	0.021
stress-test window (0.85)				
5 m $\dagger$	0.308	1.0	0.308	0.301
30 m $\dagger$	0.317	4.5	0.277	0.418
1 h	0.322	8.8	0.238	0.418
2 h	0.322	8.8	0.282	0.418
6 h	0.743	20.4	0.439	0.995
1 d	0.743	20.4	0.439	0.995

TABLE X

e-LOND true detections under pre-committed, evidence-independent within-bucket orders, all five windows, detector seed 0. All orders emit hypotheses at bucket close and differ only in how groups closing in the same bucket are sequenced. The count is materially order-sensitive, and first-flow arrival is the largest over every order and seed we audit — which is why the paper reports the key-hash row, a deterministic hash of the group’s own (SrcIP, DstIP, bucket) key (collisions broken on the key), and carries first-flow only as a labelled upper bound. The mechanism is Proposition 2: before the first rejection only a prefix of 902 / 902 / 865 / 824 / 748 steps out of 57,368 / 49,267 / 37,231 / 31,672 / 31,568 can reject at all, so a detection needs a near-ceiling episode to land inside it. Under the canonical order at 0.70 and 0.77 no episode clears its own step anywhere in the stream, which is why those entries are 0. The key-hash is a timing-independent canonicalization, not a security mechanism: a public hash of attacker-selectable fields can be ground, and the keyed row is the grinding-resistant variant, itself an unpredictable draw from the random ensemble below. Conditioned on there being detected episodes, the per-order median padding cost is 109.5 / 16 / 93 / 175 / 98 flows — a survivorship-conditioned estimand. Because the key-hash reads no arrival time, it is also what makes the padding attack leave Assumption 1 intact for every other hypothesis: injecting one pad at the earliest instant of its own bucket moves 373 / 348 / 4 / 7 / 3 other hypotheses under first-flow arrival, against 0 under the key-hash at every window and seed. The canonical order is also the more seed-stable of the two: across detector seeds its count moves by at most 3, against 30 for first-flow arrival. Feasibility (C1) is order-invariant. See section IX-D.

within-bucket order	0.55	0.62	0.70	0.77	0.85
first-flow arrival (upper bound)	18	13	30	31	72
group-id (packed key)	0	3	0	0	24
last-flow arrival	0	0	0	0	0
fixed random permutation (one instance)	2	1	9	1	38
key-hash (canonical, reported)	3	11	0	0	34
keyed key-hash (secret seed)	0	6	3	3	24
median over 50 orders	1	6	2.5	1	23.5
range over 50 orders	0–10	0–11	0–12	0–6	0–44

TABLE XI

Randomised smoothing at the primary window (LOND, $\gamma \propto j^{-1.6}$, position 0.55, seed 0): removing the floor adds no reliable detection that survives the merge choice. Rejections are mean $\pm$ SD over 100 randomisation draws where applicable; “reliable” is the number of malicious episodes detected with probability ≥ 0.9 , “any” with probability > 0 . All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

merge	route	rejections	reliable	any
mean-e	discrete	18.0	18	18
simes	smoothed	24.1 ± 2.7	18	100
hommel	smoothed	12.5 ± 4.7	1	67
bonf-p	smoothed	23.2 ± 2.7	17	100

TABLE XII

Periodic restart at the primary window (e-LOND, seed pair 0/1 — decision-identical to LOND here, since the group p-value is $\min(1, 1/\text{Ev})$ so $p \leq \alpha_t \Leftrightarrow \text{Ev} \geq 1/\alpha_t$): detection restored at the cost of a per-epoch guarantee (total spend $n \cdot q$). Budget reset (E2c) spends $\alpha_{\text{tot}} = n \cdot q$; index reset (E2b) keeps $\alpha_{\text{tot}} = q$ via a pre-committed allocation. Rejections are shown per seed (0/1); recall and FDP are the two-seed mean. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

arm	epoch (h)	α_{tot}	rej. (s0/s1)	recall	FDP
uninterrupted (baseline)	–	0.050	18 / 18	0.065	0.000
restart 2 h, budget reset (E2c)	2	0.250	99 / 91	0.345	0.000
restart 1 h, budget reset (E2c)	1	0.450	130 / 121	0.406	0.000
restart 2 h, index reset, uniform (E2b)	2	0.050	55 / 52	0.195	0.000
restart 2 h, index reset, geometric (E2b)	2	0.050	28 / 26	0.098	0.000
[oracle] uninterrupted	–	0.050	104 / 96	0.364	0.000

TABLE XIII

The UAI 2026 procedures [20] against the feasibility horizon ($\gamma \propto j^{-1.6}$, seed 0, two-hour grouping; section IV-C). “det.” is true detections, “silent” the fraction of hypotheses that cannot be rejected at their own arrival even at the evidence ceiling. For the step-up procedures that fraction is computed exactly, by re-running the step-up on the counterfactual in which the arriving hypothesis carries M ; the cheaper test — does M clear $1/(\alpha\gamma_t r)$ at the largest conceivable r — is only necessary, since reaching r needs r hypotheses to clear the bar at once, and it reports 0.0% at every window here. All counts and silence fractions are under the canonical within-bucket order, as table I is. Deferring the decision leaves at-arrival feasibility essentially unchanged — identical to $d_t = t$ at three of the five windows — and buys power only by revisiting, which is why its price is alerting latency. The last row is online e-BH, reached as the $d_t = \infty$ limit and verified rejection-identical to it at every row. Donation e-BH is the literal reading of its source’s Eq. (102); a second reading of that equation gives one more detection at 0.85 and is reported here as a sensitivity analysis, and neither is quoted as a power comparison (section IV-C).

Procedure	0.55		0.62		0.70		0.77		0.85	
	det.	silent	det.	silent	det.	silent	det.	silent	det.	silent
e-LOND (baseline), $d_t = t$	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.1%
donation e-LOND	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.1%
closed e-LOND (e-LOND)	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.0%
donation e-BH	3	—	11	—	0	—	0	—	35	—
e-TOAD, bucket-close deadline	3	96.3%	11	91.3%	0	96.5%	0	97.4%	35	77.3%
e-TOAD, $d_t = \infty$ (= online e-BH)	3	96.3%	11	91.3%	0	96.5%	0	97.4%	35	77.3%

TABLE XIV

Five padding pools priced against the running e-LOND level $1/\alpha_t$, detector seed 0, two-hour grouping, at the replication window (0.62) and the stress-test window (0.85), under both within-bucket orders — the canonical metadata hash table I reports and the first-flow arrival order kept as the optimistic upper bound. Entries are the median suppression cost r per detected episode, over the detected set that order produces (counts in the block headings); the order changes which episodes must be paid for and how deep into the level sequence they sit, which is why the two blocks differ by more than a constant. Four pools coincide within each block; service-matched is dearer, both from a larger mean e-value and from being defined on a costlier episode subset (section VI-C). μ is the pool-level mean e-value at 0.85 where defined, a property of the pool and not of the order. Protocol- and service-matched use the episode’s modal protocol/port. Costs are lower bounds.

Padding pool	μ (0.85)	median r (0.62)	median r (0.85)
canonical metadata-hash order (headline)	—	11 detected at 0.62, 34 at 0.85	—
generic benign	50.9	6	116
attacker-origin	216.8	6	116
protocol-matched	—	6	116
service-matched	—	7	175
black-box	0.0	6	116
first-flow arrival (optimistic upper bound)	—	13 detected at 0.62, 72 at 0.85	—
generic benign	50.9	72	5,212
attacker-origin	216.8	72	5,247
protocol-matched	—	72	5,232
service-matched	—	99	10,299
black-box	0.0	72	5,202

TABLE XV

Pre-committed weight schemes at the primary window, LOND, $\gamma \propto j^{-1.6}$, mean over two detector seeds. $T = 57,368$ episodes, 275 malicious, $|\mathcal{C}| = 2,448,993$. L^* is quoted at the unperturbed level and is therefore an upper bound. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else. Both legs of every comparison here use it, so the order moves the level and not the ranking.

Rule	pad-inv.	recall	L^*	appended	ratio
arithmetic mean (symmetric)	—	0.065	—	—	—
first-event-only	yes	0.062	1	111	112×
exp-decay $\rho = 0.5$	yes	0.062	2	111	56×
uniform over 10 slots	yes	0.062	8	111	14×
uniform over 100 slots	yes	0.027	81	492	6×
exp-decay $\rho = 0.99$	yes	0.027	164	492	3×
last-event-only	no	0.045	—	336	—
security-prior (out of class)	yes	0.000	—	—	—

TABLE XVI

Padding suppression cost priced against the running controller level $1/\alpha_t$ (real) vs the static T/w_0 threshold. Part (a) gives both within-bucket orders at both detector seeds: the canonical metadata-hash order the paper reports, and the first-flow arrival order retained as an optimistic upper bound over the orders we audit (table X). Detections and the level an alert fires at both move with the order; T , $|C|$, AUROC and the margin do not. Part (b)’s pool comparison is on the first-flow detected set, the larger of the two; table XIV gives the same comparison under both orders. Four pools coincide; service-matched is dearer. Medians are rounded to the nearest integer; “–” marks a window/seed/order with no detections.

(a) per window $\times$ seed $\times$ order (e-LOND poly, two-hour grouping)								
pos	seed	canonical key-hash			first-flow (upper bd.)			
		det.	pad (real)	pad (static)	det.	pad (real)	pad (static)	
0.55	0	3	24	3	18	118	4	
0.55	1	2	27	2	18	101	3	
0.62	0	11	6	4	13	72	4	
0.62	1	8	5	2	12	69	0	
0.70	0	0	–	–	30	1,851	62	
0.70	1	0	–	–	0	–	–	
0.77	0	0	–	–	31	1,165	85	
0.77	1	0	–	–	31	1,289	90	
0.85	0	34	116	34	72	5,202	33	
0.85	1	32	109	34	71	6,802	33	

(b) five padding pools, seed 0 (median suppression cost r)			
padding pool	real (0.62)	real (0.85)	static (0.85)
generic benign	72	5,212	33
attacker-origin	72	5,247	33
protocol-matched	72	5,232	33
service-matched	99	10,299	35
black-box	72	5,202	33

TABLE XVII

Aggregation-cap sweep (median over five positions $\times$ two seeds): the frozen $n_0 = p_{99}$ is chosen for validity and attack cost, not power. “% groups $m > n_0$ ” is the share of groups whose size exceeds the cap; “front-load cost” is the number of appended flows needed to defeat the front-loading defence. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

cap	n_0	% groups $m > n_0$	det. raw	det. trunc.	front-load cost
mean	47	5.72	39.5	35	16
p50	2	37.58	99	71	1
p90	21	10.24	68.5	60	8
p99	408	1.08	0	0	116
p999	7,258	0.12	0	0	3,560
max	97,911	0.01	0	0	–

TABLE XVIII

ADDIS controller-state attack: $B^* = 203$ precursor episodes permanently silence the controller at position 0.85 — the known-invalid stress window, which is where the real-stream demonstration must sit, since ADDIS’s horizon escape and its guarantee-invalidity are the same property on two-point conformal evidence (table XIX reproduces the mechanism where the guarantee does hold). The break-even is the number of detected episodes whose per-episode padding cost equals the whole one-off state attack. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

quantity	value
B^* (precursor episodes, $R = 0$)	203
per-precursor cost $\lfloor \lambda(C + 1)/k \rfloor + 1$	453,279 flows
total attack flows	92,015,637 ($\approx 9.2 \times 10^7$)
level-saturated coincidence	101 of 147
minimal pad lands in window	101 of 147
median pad per episode (alternative)	34,465,311 flows
break-even vs padding	2.67 episodes
grey-box landing band ($ \hat{C} / C $)	[0.67, 1.33]

TABLE XIX

The ADDIS state attack on a synthetic stream where ADDIS’s guarantee genuinely holds, isolating the mechanism from the real-window validity failure. The two-point conformal evidence used elsewhere cannot supply this: its ADDIS horizon-escape (the spending index never advances) is the same property that makes its null p-values non-conservative, so no two-point stream is both escape-feasible and guarantee-valid. We therefore use genuinely uniformly-conservative null p-values (Unif(0, 1), independent, the assumption ADDIS needs). The stream is synthetic, so LSPR23’s grouping plays no part and the result is order-invariant by construction. Top: ADDIS controls FDR — an expectation, not a per-realisation bound (the mixed arm’s worst single realisation is FDP = 0.104, above q , which the guarantee permits) — on this stream: the honest all-null test (every rejection false) gives mean FDP = 0.010 $\leq q=0.05$, and with 120 strong targets present mean FDP = 0.047 at recall 1.00 (200 seeds each). Bottom: the attack is adversarial state poisoning — prepending B attacker-generated alternative (non-null) precursor episodes with $p \in (\lambda, \tau]$. ADDIS constrains only true-null p-values, so alternatives may take any value: the legitimate nulls stay Unif(0, 1) and ADDIS’s guarantee holds on the attacked stream, not merely the clean one. These selected non-candidates advance the spending index and silence every rejection at a median $B^* = 140$ over 300 guarantee-valid streams (range 140–255; the tail from a rare null false positive that transiently restores wealth). So even where ADDIS’s guarantee holds after the attack, an adversary who only adds traffic drives the controller to silence — a property of the mechanism, not an artefact of the dataset already violating the assumption.

	mean FDP	$\leq q=0.05?$
all-null (every rejection false)	0.010	yes
mixed (120 targets, recall 1.00)	0.047	yes
state poisoning: median $B^*=140$ (range 140–255) silences all		

TABLE XX

Both attacks in operational units (position 0.85 — the known-invalid stress window; these are attack costs, which are measured against the controller’s level and do not rest on the evidence being a valid e-value — black-box padding pool). Wire bytes are the per-flow median; rate is priced over the relevant span (7200s window for padding, the position’s own deployment span for the state attack). “ $\times$ window” and “ $\times$ dataset” are flow-count ratios. Each padding row names the procedure whose level it is priced against, whether that level is the running $1/\alpha_t$ at the episode’s own step or the static cold-start τ , and the detected set it is summed over, because these differ by two orders of magnitude and are easily confused, and the within-bucket order the detected set was produced under. Row 1 is the headline cost: the median over e-LOND’s own 34 detections at 0.85 under the canonical order, priced at the running level — the value table I quotes. Row 2 is the same quantity over the 72 detections of the first-flow arrival order, the optimistic upper bound over the orders we audit; it is $45\times$ dearer because that order detects more episodes and detects them deeper into the level sequence, where $1/\alpha_t$ is larger. Both appear in table XVI. Rows 3–4 are the cheap extreme: r_{90} over the 257 episodes of positions 0.62 and 0.85 (two seeds) at which all five pools are defined, priced against the static level- w_0 threshold T/w_0 , first-flow. Row 5 multiplies that static median by ADDIS’s 147-episode detected set (table XVIII), so it mixes an e-LOND-scale per-episode cost with an ADDIS-scale count; it is the cheapest reading of “suppress everything” and is reported only for contrast with the last two rows. Every row below the first is under first-flow arrival.

attack budget	flows	wire bytes	rate	$\times$ window	$\times$ dataset
e-LOND, running $1/\alpha_t$: one alert (median), canonical	116	45.5 KB	50.5 bit/s	4.7×10^{-5}	7.1×10^{-6}
same, first-flow (optimistic upper bd.)	5,202	2.0 MB	2265.8 bit/s	2.1×10^{-3}	3.2×10^{-4}
level- w_0 static $\tau=T/w_0$: one alert (median)	34	13.3 KB	14.8 bit/s	1.4×10^{-5}	2.1×10^{-6}
level- w_0 static $\tau=T/w_0$: one alert (p90)	227	89.0 KB	98.9 bit/s	9.3×10^{-5}	1.4×10^{-5}
static median $\times$ ADDIS’s 147 detections (mixed)	4,998	2.0 MB	2176.9 bit/s	2.0×10^{-3}	3.1×10^{-4}
ADDIS, running $1/\alpha_t$: one episode (median)	34,465,311	13.5 GB	15.0 Mbit/s	14.05 $\times$	2.11 $\times$
ADDIS spending-state attack (B^* precursors)	92,015,637	36.1 GB	33.4 Mbit/s	37.51 $\times$	5.63 $\times$

TABLE XXI

Controlled padding-dilution on the shipped HGB detector (src-dst two-hour unit, e-LOND $\gamma \propto j^{-1.6}$, seed 0). Pad flows are real flows of the black-box pool, selected on network-observable frequency alone: the most common (proto, port) over the training prefix, which precedes both calibration and deployment — no labels, no detector output, no knowledge of the attacker’s own traffic. It picks proto 17/port 53 at both windows, the same service the label-using rule would, and the resulting pool is measured to be 0.0000% attack-labelled (an audit, not a selection criterion). The selection rule matters: taking the mode over all window traffic instead picks the attacker’s own flood at the stress window, a pool 90% attack-labelled that does not dilute. “pad fires” is how many of n sampled pad flows fire when run through the detector: 0 observed at both windows, whose exact one-sided 95% Clopper–Pearson upper bound is the next column — evidence for the structural premise (ordinary victim-service traffic scores like ordinary traffic), not proof that the firing probability is zero. “measured = closed form” counts detected episodes whose padded group e-value $E(G) = S/(m+r)$ crosses the firing threshold at exactly the predicted $r^* = \lfloor S\alpha_t \rfloor - m + 1$; the median r^* reproduces table I. Both within-bucket orders are attacked. The key-hash rows are the canonical order the paper reports — the one under which a pad provably moves no other hypothesis’s spending weight, so the Assumption 1 argument is clean; the first-flow rows re-run the identical chain — same detector, same pool, same pricing — on the detected set the initial first-flow order produces, which table X shows is the largest over the orders we audit. Every detection is suppressible under both. The canonical primary-window set is small, so read it as three individually-priced episodes (costs 23, 24, 33 flows) rather than as a rate; the canonical costs are lower than the first-flow ones because the two orders detect different episodes at different steps of the level sequence, not because the canonical order is weaker.

window	within-bucket order	pool flows	pad fires	95% upper	detected	measured = c.f.	median r^*
primary (0.55)	first-flow (upper bd.)	845,691	0/20,000	1.5×10^{-4}	18	18/18	118
	key-hash (canonical)	845,691	0/20,000	1.5×10^{-4}	3	3/3	24
stress-test (0.85)	first-flow (upper bd.)	584,850	0/20,000	1.5×10^{-4}	72	72/72	5,202
	key-hash (canonical)	584,850	0/20,000	1.5×10^{-4}	34	34/34	116

TABLE XXII

Surface A priced without oracle state: a re-analysis of the same per-episode (m, S, α_t) the oracle costs come from, with no re-simulation — the closed form (??) is re-derived from the stored fields on all 127 episodes as a read-back check. Multiplier: append $(c-1)m$ flows, that is, multiply your own episode by c . It defeats the alert iff $c \geq 1 + r^*/m$ and needs no estimate of S, α_t or $|C|$ — only m , which on LSPR23 is the attacker’s own flow count, since every host pair carrying attack traffic is 100% malicious. “padded arity” is cm , and the column beside it answers the deployment-side question — would a volume monitor see the pad — as the fraction of deployment episodes larger than the padded one, bracketed by the two stored group-size quantiles that straddle it. Those quantiles are not shown and are not benign deployment values: they are quantiles of source–destination group size over the calibration slice, unfiltered by label, which is why the deployment exceedance and not the quantile is what the claim rests on. Rows are given under both within-bucket orders — the canonical metadata hash the paper reports and first-flow arrival, retained as the optimistic upper bound. Absolute: append a pre-committed flat N flows, which needs nothing at all; N is reported at the fraction of that cell’s alerts it suppresses. The multiplier arm covers the two windows for which per-episode arities are stored; the replication window appears in the absolute arm only. Both strategies are upper bounds on the oracle cost, and both are optimistic on a benign-inclusive stream, where m also contains benign flows the attacker cannot see.

multiplier — needs the attacker’s own arity m and nothing else									
order	pos	alerts	med. c	max c	c used	med. added	vs oracle	padded arity	% deployment larger
canonical	0.55	3	1.57	2.74	3	80	3.5×	120	0.9–9.9%
canonical	0.85	34	2.76	6.17	10	792	5.1×	880	0.1–1.1%
first-flow	0.55	18	3.54	7.44	10	414	3.6×	460	0.1–0.9%
first-flow	0.85	72	73.19	4302.15	none ≤ 100	—	—	—	—

absolute — a pre-committed flat pad, needs no knowledge at all						
order	pos	seed	alerts	N for 50%	N for 90%	N for all
canonical	0.55	0	3	24	33	33
canonical	0.55	1	2	30	30	30
canonical	0.62	0	11	6	17	859
canonical	0.62	1	8	6	659	659
canonical	0.85	0	34	118	9,395	58,616
canonical	0.85	1	32	111	9,426	58,620
first-flow	0.55	0	18	125	645	837
first-flow	0.55	1	18	108	645	829
first-flow	0.62	0	13	72	127	134
first-flow	0.62	1	12	70	93	98
first-flow	0.77	0	31	1,165	3,736	7,886
first-flow	0.77	1	31	1,289	4,258	7,886
first-flow	0.70	0	30	1,877	4,145	5,445
first-flow	0.85	0	72	6,714	69,195,773	145,525,527
first-flow	0.85	1	71	6,802	69,086,348	145,096,310

TABLE XXIII

Host-conditioned detector and the padding-transfer boundary (src-dst two-hour unit, e-LOND $\gamma \propto j^{-1.6}$, seed 0). The detector reads the 33 flow features plus six strictly-causal ($ts < ts(\text{flow})$) host-context features — per-host prior counts, distinct-peer counts, and failed-connection fractions — and no endpoint ground-truth (label_src/label_dst both have attack-rate 1.0, excluded). It has higher AUROC than the flow-only detector at both windows, yet lower tail reach (the fraction of attack flows above all benign calibration scores) and fewer e-LOND detections (2 vs 18 at the primary window) — higher ranking quality buying fewer, not more, operational alerts, the detector-quality-vs-feasibility gap of C1. To test the padding boundary we graft each attacked pair’s real causal context (strongest feature src-distinct-dst, $\rho = +0.34$; padding one victim cannot raise it) onto ordinary black-box pads — the most common (proto, port) on the training prefix, which precedes both calibration and deployment, so the pool is chosen on network-observable frequency alone with no labels and no detector output — and replay the append causally. The pads fire zero across accumulation at both windows (over 4×10^6 premise trials at the primary window), and the suppression cost is the identical flow-level closed form ($0.55 r^* = 108$ host vs 118 flow; $0.85 r^* = 11,484$ host vs 5,202 flow), so the transfer holds at both. A non-black-box pool rule does fire: selecting the modal service among detector-non-firing flows — picks TCP/80 at 0.85, a pool that is only 12% benign and fires 0.42. Grafting the attacker’s own flood onto attack context and observing the detector fire is not evidence of a defence, so we report that rule only as a sensitivity. What LSPR23 still cannot settle is the benign-inclusive question at all: its attacked pairs are 100% malicious, so no real ordinary-to-victim traffic exists here — which is what the AIT testbed (tables XXIV and XXV) supplies. “real benign fires” is host detector firings among the window’s benign flows; “graft fire” is the grafted-pad fire rate on detected episodes. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

window	AUROC h/f	tail reach h/f	det. h/f	real benign fires	graft fire	transfer
primary (0.55)	0.954/0.916	0.009/0.033	2/18	0/2,287,988	0.000	holds
stress-test (0.85)	1.000/0.999	0.295/0.610	37/72	3/1,638,722	0.000	holds

TABLE XXIV

The padding-transfer boundary re-measured on a benign-inclusive testbed (AIT-LDSv2.0, 8 organisations, 45,001 attack flows), where victims serve heavy real benign traffic while attacked, so the padding question is answered on real ordinary-to-victim flows — no graft. Leave-one-organisation-out (train 7, test 1; a “deploy on a new network” test), in-organisation benign conformal calibration, flow features selected per fold from the training orgs with all identity and absolute-time fields excluded. “benign→victim” is the fire rate of real benign flows to an attacked host (the pad analogue), host / flow-only. Host conditioning adds little and inconsistently (median host–flow delta -0.0000 ; host higher on 1/8 orgs, max 8.9%), with shaw the one organisation where it is materially higher. That per-organisation exception is real but does not survive as a defence. A raised ordinary-to-victim fire rate is measured here in each flow’s own context; the pads an attacker actually sends carry the attacked pair’s context, and replaying the whole Surface A chain under that context suppresses every detected Shaw episode (table XXV). So the honest reading is that host conditioning is usually negligible here and can look decisive on an individual organisation, but on the two organisations we replay end-to-end it does not stop the attack once the pad’s own context is modelled — nine episodes, not a general claim about host-conditioned detectors. This also settles the LSPR23 side: the 0.43 grafted firing at the stress window comes from a non-black-box pad pool that is mostly the attacker’s own flood (table XXIII). AIT has no high-volume flood, which remains open. The flow-only folds are audited under both within-bucket orders in table XXVI: the canonical order detects fewer episodes and its alerts are cheaper to suppress, as on LSPR23. Rows sequenced by first-flow arrival are conditional on the detected set that order produces on this testbed.

organisation	AUROC h/f	atk. recall	benign→victim % (h/f)
fox	0.995/0.996	0.93	0.00/0.01
harrison	0.975/0.973	0.86	0.01/0.01
russellmit	0.988/1.000	0.47	0.00/0.02
santos	0.948/0.980	0.36	0.00/0.01
shaw	0.955/0.932	0.47	8.92/0.05
wardbeck	0.969/0.907	0.39	0.14/0.01
wheeler	0.999/0.998	0.97	0.01/0.01
wilson	0.974/0.984	0.85	0.07/0.00

TABLE XXV

Full Surface A suppression pipeline on the benign-inclusive AIT testbed, with real ordinary-to-victim pads replayed against the running controller level (no graft). Per organisation: a leave-one-organisation-out detector whose flow features are chosen per fold from the training organisations only, in-organisation benign conformal calibration on flows strictly preceding the first attack, (SrcIP, DstIP, 2 h) episodes, e-LOND detection, then each detected malicious episode suppressed by replaying real benign flows to an attacked victim of the organisation (200 draws). A firing pad contributes the maximum e-value and raises the cost, which the zero-pad closed form misses. The two arms replay differently because the two detectors read different things. Flow-only: flow features carry no context, so pads are drawn with replacement from the pool’s observed 0-or- M evidence — an empirical i.i.d. generator, so a modelled cost may exceed the finite observed pool. Host-conditioned: the six host features are causal, so appending pads changes the context every later pad is scored in; holding them fixed would be a static-context diagnostic rather than a replay. We therefore recompute the victim’s context as pads accumulate (counts raised, distinct-peer counts pinned, failure fractions moved to the pool’s rate) and score the whole pool under it, reporting a success only over the accumulation range actually scored. “det.” is e-LOND true detections; “pad fire” the real ordinary-to-victim firing rate as observed; “ r^* emp.” the empirical median cost over suppressible episodes; “ r^* closed” the zero-pad closed-form lower bound over the same suppressible episodes ($\leq$ empirical by construction); “suppr.” the number of detected episodes suppressible within budget; “succ.” the mean per-draw suppression success. The chain runs end-to-end at all eight organisations: 84 of 85 detections are suppressible, per-draw success 0.80–1.00. Host conditioning does not stop it either, and the causal replay is what shows that: Shaw’s ordinary-to-victim flows fire 7% in their own contexts, which under a static-context replay made no episode suppressible — but a pad’s context is the attacked pair’s, and under that context the same pool fires at most 0.5% across accumulation, so all three episodes suppress. The pinning models a single already-active attacker–victim pair; padding from new hosts would move the distinct-peer counts and is not covered. The flow-only folds are audited under both within-bucket orders in table XXVI: the canonical order detects fewer episodes and its alerts are cheaper to suppress, as on LSPR23. Rows sequenced by first-flow arrival are conditional on the detected set that order produces on this testbed.

org	det.	pad fire	r^* emp.	r^* closed	suppr.	succ.
flow-only detector						
fox	13/14	1.3e-04	277	277	13/13	1.00
harrison	10/14	1.8e-04	1,412.5	1,412.5	10/10	0.99
russellmitchell	14/14	5.2e-04	3,551.5	3,365.5	14/14	0.96
santos	9/9	4.7e-04	212	212	8/9	0.80
shaw	3/6	0.0e+00	404	404	3/3	1.00
wardbeck	9/9	3.2e-04	8,294	8,294	9/9	0.90
wheeler	15/16	2.1e-04	1,609	1,609	15/15	1.00
wilson	12/13	4.3e-05	793	793	12/12	1.00
host-conditioned detector, end-to-end						
shaw	3/6	7.0e-02	269	269	3/3	1.00
wilson	6/13	0.0e+00	364.5	355	6/6	1.00

TABLE XXVI

The AIT transfer under both within-bucket orders, flow-only folds.

This is a resequencing; the leave-one-organisation-out feature selection, the per-fold detector fit, the scores, the chronological in-organisation calibration and the e-values are all computed before episodes are ordered, so the two arms differ by the permutation alone; the canonical arm uses the same key-hash rule as the LSPR23 results. The stored first-flow arm is re-derived here as a read-back check that this is the same chain and not merely a similar one (it reproduces 84/85). The canonical order detects fewer episodes, as on LSPR23, and the alerts it does issue are cheaper to suppress — per-organisation median r^* falls from 212–8294 to 12–4650 flows. The host-conditioned arm is two organisations and nine episodes and is left on first-flow (table XXIV). “det./mal.” counts detected episodes against malicious episodes; costs are oracle lower bounds.

organisation	det./mal.	median r^*	suppressible	per-draw succ.
canonical order — 78/79 suppressible				
fox	11/14	12	11/11	1.00
harrison	9/14	50	9/9	1.00
russellmitchell	14/14	4,650	13/14	0.93
santos	8/9	35.5	8/8	1.00
shaw	3/6	26	3/3	1.00
wardbeck	8/9	65	8/8	1.00
wheeler	15/16	192	15/15	1.00
wilson	11/13	274	11/11	1.00
first-flow order — 84/85 suppressible				
fox	13/14	277	13/13	1.00
harrison	10/14	1412.5	10/10	0.99
russellmitchell	14/14	3551.5	14/14	0.96
santos	9/9	212	8/9	0.80
shaw	3/6	404	3/3	1.00
wardbeck	9/9	8,294	9/9	0.90
wheeler	15/16	1,609	15/15	1.00
wilson	12/13	793	12/12	1.00

TABLE XXVII

Matched operating points at the primary analysis window (position 0.55, two-hour grouping, $T = 57,368$ episodes of which 275 are malicious; every FDR statement conditional on Assumption 1 and Lemma 1). Rows are under the canonical within-bucket order, as table I is, with the first-flow arm — the optimistic upper bound — beneath for the four rows the order moves. The stress-test window 0.85 is table XXVIII; the negative result, every method within 0.047 recall of the frontier, holds at both and under every order audited. The frontier is an ex-post oracle (thresholds use the test labels) and is measured order-invariant here, so it is quoted once. The slot rule is an expectation over slots and the no-feedback threshold never updates, so both are order-free by construction and verified unchanged.

Method	alerts	FDP	recall	gap
canonical metadata-hash order				
online FDR (mean rule)	3	0.000	0.011	+0.000
online FDR (slot rule)	91.0	0.000	0.331	+0.000
feedback, $L=0$	117	0.068	0.396	+0.000
feedback, $L=20$	137	0.204	0.396	+0.015
feedback, $L=50$	163	0.331	0.396	+0.044
fixed threshold	1,047	0.836	0.625	+0.000
first-flow arrival (optimistic upper bound; unmoved rows omitted)				
online FDR (mean rule)	18	0.000	0.065	+0.000
feedback, $L=0$	119	0.076	0.400	−0.004
feedback, $L=20$	135	0.185	0.400	+0.004
feedback, $L=50$	150	0.267	0.400	+0.040
frontier, FDP=0.000	104	0.000	0.378	—
frontier, FDP=0.050	110	0.009	0.396	—

TABLE XXVIII

Matched operating points at the stress-test window (position 0.85, two-hour grouping, $T = 31,568$ episodes of which 255 are malicious). FDP is measured against labels at a window where the evidence is not a valid e-value (section IX-A). Rows are under the canonical within-bucket order, with the first-flow arm beneath for the 4 rows the order moves; the frontier is measured order-invariant and quoted once. The companion primary-window table is table XXVII.

Method	alerts	FDP	recall	gap
canonical metadata-hash order				
online FDR (mean rule)	34	0.000	0.133	+0.000
online FDR (slot rule)	126.6	0.001	0.496	−0.003
feedback, $L=0$	160	0.069	0.584	+0.004
feedback, $L=20$	171	0.117	0.592	+0.024
feedback, $L=50$	190	0.184	0.608	+0.047
fixed threshold	875	0.763	0.812	+0.000
first-flow arrival (optimistic upper bound; unmoved rows omitted)				
online FDR (mean rule)	72	0.000	0.282	+0.000
feedback, $L=0$	161	0.068	0.588	+0.000
feedback, $L=20$	160	0.081	0.576	+0.012
feedback, $L=50$	171	0.140	0.576	+0.039
frontier, FDP=0.000	117	0.000	0.459	—
frontier, FDP=0.050	156	0.045	0.584	—

TABLE XXIX

Analyst-feedback controllers under wall-clock disposition delay ($q = 0.05$) at the primary analysis window (0.55); FDP is the two-seed mean. “open-loop” and “at limit” are for the P controller (open-loop is identical for PI). The final row is the long-span window at a 24 h delay, where feedback collapses to the no-feedback open-loop baseline. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

delay (h)	FDP (P)	FDP (PI)	FDP (AQT)	% open-loop	% at limit
0	0.053	0.049	0.172	0.9%	95.0%
0.25	0.402	0.402	0.523	51.4%	87.3%
1	0.514	0.514	0.592	60.5%	73.4%
4	0.671	0.671	0.671	77.8%	44.7%
8	0.834	0.834	0.834	99.8%	12.0%
long-span, 24 h	0.877	0.877	0.876	no-feedback FDP = 0.887	

TABLE XXX

Benign firing rate vs nominal by window (seed 0). “ratio” is the measured/nominal benign firing rate at $k = 1$, and “95% CI” is its exact Clopper–Pearson binomial interval given the observed firing count out of the benign flows. The interval separates the theoretical guarantee from the empirical diagnostic: at the four non-stress windows only 0–3 benign flows fire (expectation ≈ 1), so the interval is wide and includes 1 — the diagnostic cannot prove validity, only fail to reject it; at position 0.85 the interval excludes 1 by orders of magnitude, a clear violation. The last four columns are the ratio at rank depth $k \in \{1, 10, 100, 1000\}$. This diagnostic is order-invariant: it scores benign flows against the calibration threshold, with no controller and so no stream sequence to permute.

position	AUROC	fired benign	ratio	95% CI	ratio at rank depth k			
					$k=1$	$k=10$	$k=100$	$k=1000$
0.55	0.916	1	1.07	[0.03, 5.96]	1.07	0.64	0.51	3.85
0.62	0.799	1	1.16	[0.03, 6.45]	1.16	7.17	0.91	5.09
0.70	0.914	3	3.79	[0.78, 11.06]	3.79	0.50	0.09	0.11
0.77	0.828	2	2.72	[0.33, 9.82]	2.72	0.95	0.98	2.41
0.85	0.999	46	50.90	[37.26, 67.89]	50.90	9.18	4.49	6.80

TABLE XXXI

Metadata-stratified interrogation of Assumption 1 (seed 0). The assumption constrains benign flows of true-null groups conditionally on that group’s metadata, but our other diagnostic (table XXX) reads the marginal rate at the shipped depth $k=1$, where the four non-stress windows fire 0–3 benign flows. Here those flows are stratified on the observable components of $\mathcal{M}_j$ — episode arity (the quantity padding manipulates), service (proto, port), the two-hour bucket that fixes a group’s position, and high-support destination host, all fixed by an outcome-blind support rule — and read at $k \in \{1, 10, 100, 1000\}$. Both sources of uncertainty are resampled jointly, and both are clustered: a window’s whole diagnostic depends on the calibration set only through its k -th largest score, which every stratum shares, so the calibration tail is resampled over calibration host pairs; the test side is a Poisson-multiplier bootstrap over deployment host pairs. Ranks break ties conservatively, so each ratio is a lower bound on the anti-conservatism. Multiplicity is corrected with Benjamini–Yekutieli, valid under the arbitrary dependence this design has, over every cell: a cell firing fewer than ten times carries no test, but still enters the family at $p = 1$, since dropping it would shrink the denominator using the same tail count that drives significance. “contrast” counts strata departing from their own window’s marginal on the same replicates — the quantity that speaks to the conditional statement, since the shared calibration draw largely cancels in a ratio of two rates read at one threshold. Reading, in both directions. 428 of the 536 cells carry no test at all (419 of them fire fewer than ten times), which is itself the point: at the depth the pipeline uses, the data mostly cannot speak. Of the 40 cells that do reject, none at $k=1$ rejects at the primary or replication window — all 5 $k=1$ rejections sit at 0.85, already an unambiguous violation, which is an inconclusive non-rejection and not evidence for Assumption 1: at $k=1$ most cells are simply uninformative. Deeper in the tail the conditional statement is another matter: 32 stratum contrasts survive correction, at windows 0.55, 0.62, 0.70, 0.77, 0.85. The largest lower bound is $m=21$ -100 at 0.77, $k=1000$ ($9.3 \times [6.5, 13.7]$); the largest point estimate is proto 6/port 6444 at 0.62, $k=10$ ($14.4 \times$). The arity family — the component of $\mathcal{M}_j$ the padding attack moves — carries 3 of them, the strongest being $m=21$ -100 at window 0.77, $k=1000$, firing $9.3 \times [6.5, 13.7]$ its own window’s rate. One dependence remains unresampled: host-pair clustering does not absorb a shock shared across many pairs at once, and a window’s calibration tail concentrates in its two to four buckets, so these counts may be optimistic — dropping the one stratum family aligned with that confounder leaves 25 contrasts at windows 0.55, 0.62, 0.70, 0.77, 0.85. Like table XXX, these counts are order-invariant: they are per-flow firings against a fixed threshold, not controller decisions. A departure at $k=1000$ is nonetheless compatible with a valid rank-1 rate — the excess may sit entirely in ranks 2..1000 — so this tests a necessary implication where it is estimable, not the assumption itself. Every guarantee-bearing conclusion in this paper is therefore stated under Assumption 1, and this diagnostic is not offered as support for it.

position	strata	cells		surviving BY			marginal ratio [95% CI]	
		total	untested	ratio	at $k=1$	contrast	$k=1$	$k=1000$
0.55	28	112	90	6	0	5	1.07 [0.00, 4.56]	3.85 [1.69, 8.70]
0.62	28	112	82	13	0	9	1.16 [0.00, 94.53]	5.09 [1.74, 11.26]
0.70	26	104	98	0	0	3	3.79 [0.00, 9.02]	0.11 [0.04, 0.28]
0.77	26	104	86	5	0	9	2.72 [0.00, 12.90]	2.41 [1.45, 3.80]
0.85	26	104	72	16	5	6	50.90 [3.70, 204.68]	6.80 [3.38, 13.04]

TABLE XXXII

What calibrating the grouped unit costs (seed 0; section IX-B). Left: the shipped construction, calibrated on benign flows. Right: split conformal on the group, same key and protocol, which would need only calibration/test group exchangeability. The evidence ceiling falls by the flows-per-group ratio ($40\text{--}49\times$) with T unchanged, so the margin collapses and detection at the primary and replication windows goes to zero. That is the feasibility boundary, not a weak statistic: the cold-start window is tens of hypotheses and the first firing group arrives hundreds of positions later (at 0.55, 110 groups fire and none lands inside the window); feasible-step counts match $\lfloor (\alpha M(R+1)/\zeta(1.6))^{1/1.6} \rfloor$ at every configuration. The last two columns qualify the repair without refuting it: the max statistic — the padding-robust one — is strongly arity-dependent, and arity is what the adversary controls, so group exchangeability may be fragile under group-formation shift, though the correlation alone neither refutes it nor shows that it fails; the mean is arity-invariant to within noise and is exactly the rule Theorem 10 defeats. Position 0.85 is the stress window and its detections are not guarantee-bearing (section IX-A). The flow-calibration columns use the first-flow within-bucket order (the optimistic upper bound), not the canonical order of table I; the group arm matches its tie-break to the same order, so the two remain comparable (table X).

Position	flow calibration			group calibration						arity corr.	
	$ C $	margin	det.	$ C $	margin	det. max	det. mean	cold start	1st fire	max	mean
0.55	2,448,993	+0.067	18	52,279	−0.977	0	0	81	401	+0.35	+0.04
0.62	2,449,031	+0.243	13	53,897	−0.973	0	0	83	349	+0.36	+0.01
0.70	2,287,988	+0.536	30	57,093	−0.962	1	0	86	6	+0.23	−0.12
0.77	2,116,418	+0.671	31	48,869	−0.961	2	1	78	11	+0.34	−0.01
0.85	1,813,113	+0.436	72	36,944	−0.971	42	42	65	4	+0.30	−0.07

TABLE XXXIII

An external denominator for the resolution cost (item R4). The 5-minute atom is our proxy for one attack action, so we count instead how many distinct red-team steps each issued alert merges, using LSPR23’s own task record (288 tasks, 295 timestamped submissions, 2023-03-09 07:03–15:08 UTC), which is an attacker-defined decomposition fixed before the dataset was published. A step is attributed to an alert when its submission falls in the alert’s bucket span and its task’s declared network segments meet the alert’s, or one of its compromise-report IPs is an alert endpoint. The link is the segment vocabulary — the task record’s `bt_baf_int` is the dataset’s own per-flow `baf_int` — corroborated against the machine-readable compromise reports (36/39 IPv4 appear as a flow endpoint; 83 agree, 0 disagree). Two nulls are run. The label null permutes which task each submission belongs to, holding the times fixed — it asks whether which task acted is informative, and goes degenerate at the daily bucket where every submission shares one bucket. The shift null rotates the whole red-team timeline against the stream, preserving task identity, ordering, attributes and inter-arrival structure — it asks the sharper question, whether the alignment in time is informative, and does not degenerate. $\checkmark$ marks a count exceeding that null’s 97.5th percentile. “flow” re-prices the same rule against the episode’s actual first-to-last flow span instead of its bucket span. The result is agreement in direction without discrimination. The external count and the proxy rise together (lower panel), so the resolution cost survives substituting an attacker-defined unit for ours; but the count clears the label null at only 6 of 19 cells and the shift null at only 1, so the rise is driven by a wider bucket capturing more submissions rather than by alert-level correspondence, and we report it as agreement rather than as confirmation. The record also sees the alerts at only 3 of 10 window-order cells: at 0.55, 0.62 the issued alerts sit in the feasible prefix — the first minutes of a deployment window — which predates the exercise’s first submission, so under 10% of them overlap it in time at all; at 0.85 they do overlap but only 2 of 17 tasks with a timed step declare a segment or file a compromise report. Three further limits. (i) At the 86,400s bucket both nulls tie the observation exactly: every submission shares one daily bucket, so no permutation and no rotation can move a step between alerts. (The spatial filter is not vacuous there — at 0.70/first-flow it admits 12 of 30 temporally overlapping alerts — it is the nulls that lose their power.) Only sub-daily widths inform. (ii) The whole positive result rests on six coarse segment labels: across every row the compromise-IP conjunct adds no alert the segment conjunct did not already admit. (iii) Every attributed alert sits on a labelled-malicious episode and the labels came from this same exercise, so nothing here is label-independent evidence that an alert is a true positive. (iv) A submission is when the red team reported: the lag, estimable from the compromise reports’ own timestamps, has median 170s (IQR 115–279s), small against every bucket width. Both within-bucket orders are reported: the canonical metadata key-hash table I uses, and first-flow arrival, the optimistic upper bound over the orders we audit (table X).

pos	order	bucket (s)	attrib.	steps/alert		label null		shift null		5-min atoms/alert
				bucket	flow	mean	>	mean	>	
0.55	first-flow	86400	12	18.33	18.33	18.33	–	18.33	–	38.47
0.62	key-hash	86400	1	35.00	35.00	35.00	–	35.00	–	35.00
0.62	first-flow	86400	6	41.00	41.00	41.00	–	41.00	–	32.00
0.70	key-hash	3600	35	4.29	3.13	2.59	$\checkmark$	2.02	–	1.00
0.70	key-hash	1800	7	12.57	12.57	11.06	–	5.83	–	3.00
0.70	key-hash	3600	2	18.50	12.00	14.81	$\checkmark$	11.75	–	2.43
0.70	key-hash	86400	1	20.00	20.00	20.00	–	20.00	–	29.00
0.70	first-flow	3600	32	4.31	3.18	2.63	$\checkmark$	1.99	–	1.00
0.70	first-flow	1800	13	13.23	12.62	11.65	–	6.09	–	3.51
0.70	first-flow	3600	13	18.77	17.38	15.06	$\checkmark$	11.95	$\checkmark$	6.51
0.70	first-flow	7200	12	19.08	18.67	15.47	$\checkmark$	11.45	–	13.00
0.70	first-flow	21600	12	19.08	18.67	15.28	$\checkmark$	11.55	–	13.00
0.70	first-flow	86400	12	27.00	27.00	27.00	–	27.00	–	28.33
0.77	key-hash	3600	4	4.00	4.00	3.19	–	1.68	–	10.00
0.77	key-hash	86400	2	3.00	1.00	3.00	–	3.00	–	18.00
0.77	first-flow	3600	12	3.00	3.00	2.74	–	1.32	–	9.93
0.77	first-flow	7200	13	3.15	3.15	3.17	–	2.94	–	19.61
0.77	first-flow	21600	13	3.15	3.15	3.15	–	3.15	–	21.61
0.77	first-flow	86400	13	3.15	3.15	3.15	–	3.15	–	21.61

(b) within-window rank agreement across bucket widths

pos	order	widths	steps vs. atoms	steps vs. bucket	atoms vs. bucket
0.7	key-hash	4	+0.800	+1.000	+0.800
0.7	first-flow	6	+1.000	+0.986	+0.986
0.77	first-flow	4	+0.816	+0.775	+0.949

TABLE XXXIV

What the results look like at lower base rates (item R5). LSPR23 is a live-fire exercise: 10.06% of flows and 0.48–0.81% of episodes are malicious, which is the optimistic case. We therefore treat prevalence as a swept parameter rather than asserting a figure for a typical SOC: π runs two to three orders of magnitude below the exercise, and the targets are a sensitivity range, not a claim about any deployment. We thin malicious episodes — not flows, which would change group arity and evidence sums and so confound C1 with C2 — to each target π , keeping a nested random subset so a lower target keeps a strict subset of a higher one. The primary mechanism replaces each thinned episode with benign evidence drawn from the same window, so the hypothesis stays at its own index and T , the margin and every survivor’s level are held exactly fixed; the deletion mechanism removes it instead, which shrinks T by up to 0.80% and raises the margin by up to +0.0129. The two agree to within 0.02 in bootstrap probability, so it is prevalence and not re-indexing that drives the collapse. “ R ” is mean rejections over 50 draws and “boot” the fraction of draws in which the controller ever rejects; the observed and $\pi = 0$ arms are single deterministic streams. $\pi = 10^{-2}$ is unreachable — it is above the observed episode rate, and we cannot manufacture attacks. Feasibility is untouched while detection collapses, which is the paper’s feasibility-is-not-detection separation driven by base rate alone. The $\pi = 0$ arm produces no rejection on any stream, but that is 5 effective (overlapping) windows, and zero would be the likely outcome even at the 4.5% early-false rate section IV quotes from simulation ($P = 0.79$): it is consistency, not a test of that number. The 0.85 rows are the stress window, whose evidence is not a valid e-value. Both within-bucket orders are reported: the canonical metadata key-hash table I uses, and first-flow arrival, the optimistic upper bound over the orders we audit (table X).

pos	order	observed		$\pi = 10^{-3}$		$\pi = 10^{-4}$		$\pi = 10^{-5}$		$\pi = 0$	
		R	boot	R	boot	R	boot	R	boot	R	boot
0.55	key-hash	3.00	100	0.54	38	0.06	4	0.02	2	0.00	0
0.55	first-flow	18.00	100	1.40	56	0.08	6	0.00	0	0.00	0
0.62	key-hash	11.00	100	0.18	14	0.02	2	0.02	2	0.00	0
0.62	first-flow	13.00	100	1.36	62	0.20	18	0.04	4	0.00	0
0.70	key-hash	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0
0.70	first-flow	30.00	100	3.52	92	0.38	28	0.10	10	0.00	0
0.77	key-hash	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0
0.77	first-flow	31.00	100	2.22	70	0.06	6	0.04	4	0.00	0
0.85	key-hash	34.00	100	0.48	26	0.02	2	0.00	0	0.00	0
0.85	first-flow	72.00	100	8.28	100	0.54	40	0.28	28	0.00	0

bootstrap probability (%) at $\pi = 10^{-4}$: the escapes collapse too					
pos	order	e-LOND	online e-BH	e-TOAD	e-LOND (deletion)
0.55	key-hash	4	4	5	4
0.55	first-flow	6	6	5	6
0.62	key-hash	2	2	0	2
0.62	first-flow	18	18	25	18
0.70	key-hash	0	0	0	0
0.70	first-flow	28	28	40	30
0.77	key-hash	0	0	0	0
0.77	first-flow	6	6	5	8
0.85	key-hash	2	2	5	2
0.85	first-flow	40	40	40	40

TABLE XXXV

Suppression by insertion, not by padding. Theorem 10 and every cost in section VI-C concern appending flows to a fixed hypothesis.

This attack appends nothing: it creates hypotheses ahead of the target, pushing it out of the cold-start window past which no rejection is possible. Two models are kept apart because they are different attacks. Targeted inserts only immediately ahead of one episode, which leaves every earlier hypothesis and hence R untouched, so its cost is closed-form

$G^* = \lfloor (\alpha(R+1)E/\zeta(1.6))^{1/1.6} \rfloor - p$ (replay-verified). Window inserts ahead of everything and silences the controller outright — that is not a per-episode cost, since it also deletes the earlier detections that raised R . The split verdict is the result: insertion does not beat padding for one alert (0/12 cells), but silences the whole window for a few hundred hypotheses (10/12 cells at one flow per inserted group, 7/12 at two). Padding totals are upper bounds — the joint cost is lower, because suppressing an early rejection lowers R — and the largest ratio is heavy-tail driven.

Panel (b) is the construction of section IX-B, which padding provably cannot suppress: calibrating on groups shrinks the ceiling, so its cold-start window collapses to 65–86 and every detection falls to a few dozen insertions. Group validity by design plus padding robustness does not buy attack resistance. Getting ahead of a target is free under first-flow arrival (send earlier in the bucket), costs an offline key-space search under a public hash, and cannot be ranked offline at all under a keyed one — which removes the search, not the attack. Both within-bucket orders are reported: the canonical metadata key-hash table I uses, and first-flow arrival, the optimistic upper bound over the orders we audit (table X).

(a) the shipped flow-level pipeline							
pos	order	det.	per episode		silence the window		cheaper
			insert	pad (flows)	insert	pad (upper)	
0.55	key-hash	3	434	24.0	217	80	–
0.55	first-flow	18	1,696	118.0	502	4,432	✓
0.62	key-hash	11	605	6.0	813	926	✓
0.62	keyed	6	452	14.5	506	126	–
0.62	first-flow	13	2,284	72.0	554	1,060	✓
0.70	keyed	3	407	97.0	257	310	✓
0.70	first-flow	30	4,088	1,850.5	706	62,250	✓
0.77	keyed	3	556	323.0	416	1,205	✓
0.77	first-flow	31	3,161	1,165.0	789	49,957	✓
0.85	key-hash	34	1,414	115.5	509	151,390	✓
0.85	keyed	24	998	160.5	518	162,496	✓
0.85	first-flow	72	5,674	5,201.5	743	786,257,264	✓

(b) the group-calibrated max pipeline — padding cannot suppress it at all							
pos	groups	firing	detections	cold start	targeted G^*	window G	
0.55	57,368	110	0	81	–	–	
0.62	49,267	109	0	83	–	–	
0.70	37,231	69	1	86	81.0	81	
0.77	31,672	60	2	78	76.5	68	
0.85	31,568	152	42	65	207.0	62	

TABLE XXXVI

What a secret canonical-order seed costs the insertion attacker. Under a public hash the adversary ranks candidate keys offline for free and instantiates only the G^* that land ahead of the target.

Under a keyed hash it cannot rank at all and must instantiate blind: $N_{50/90/99}$ is the number of distinct (SrcIP, DstIP) pairs it must actually send traffic between to suppress the target with that probability, over 400 draws. Each row's G^* is the cost on that same stream to an adversary who does know the ranking, so the last column isolates the seed and nothing else. u is the target's own hash quantile, unknown to a blind adversary; the cost scales as G^*/u , which is why the spread is wide and heavy-tailed. Every one of the 84 targets falls in EVERY draw at a budget of 1,000,000 keys — the seed does not stop the attack — but it multiplies the instantiated-key cost, at matched reliability (N_{99} against certain public suppression), by a median 19.3× (range 2.1–89.1×). Each N is located on a six-points-per-decade grid, so it is exact to within 1.48× and no further. The cost is a capability assumption, not a compute budget: N counts endpoint pairs the adversary must be able to originate real traffic between, so an adversary with only a few hundred reachable pairs is stopped outright rather than merely charged more, and these figures do not cover that regime. The model also assumes the target sits in the first bucket of its window, which holds at every cell measured; where an earlier bucket exists, insertion there precedes the target with probability 1 and the seed buys nothing. Suppression is verified by replay at every G^* , not asserted from a closed form, and the replay is checked against the binomial lower bound $P(\text{Bin}(N, u) \geq G^*_{\text{targeted}})$ it must dominate.

Rows with no detection carry no cost claim. This buys nothing against Surface A padding (Theorem 10), which needs no key at all.

pos	order	targets	u	public G^*	keyed: instantiated keys			blind N_{99}/G^*
					N_{50}	N_{90}	N_{99}	
0.55	public	3	0.037–0.072	217–217	5,663	6,639	8,962	41.3×
0.55	keyed	0	–	–	–	–	–	–
0.62	public	11	0.016–0.652	70–813	825	962	996	4.6×
0.62	keyed	6	0.060–0.413	288–506	4,994	5,843	6,053	12.3×
0.70	public	0	–	–	–	–	–	–
0.70	keyed	3	0.032–0.059	257–257	8,238	9,620	9,961	38.8×
0.77	public	0	–	–	–	–	–	–
0.77	keyed	3	0.016–0.037	416–416	26,101	30,432	31,502	75.7×
0.85	public	34	0.019–0.489	30–509	8,533	11,118	12,625	24.8×
0.85	keyed	24	0.016–0.385	78–518	8,262	9,640	9,981	19.3×

TABLE XXXVII

Adjudicated alert audit against the red-team task record (position 0.85, the known-invalid stress window). Table XXX is what establishes that invalidity, from the benign firing rate; this is a follow-up label-quality audit at the same window, not a guarantee-bearing result and not the diagnostic that made the call. Panel (a): adjudicated verdict counts with their label split. Panel (b): false-discovery proportion by method under the label, the pre-registered audit lower bound, and the strict lower/upper bounds. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else.

(a) adjudicated verdicts				
verdict	n	labelled mal.	labelled ben.	
clearly malicious	49	46	3	
probably malicious	102	100	2	
ambiguous	0	0	0	
probably benign	1	1	0	
clearly benign	0	0	0	

(b) false-discovery proportion					
method	R	FDP _{label}	FDP _{audit} ^{lo}	FDP _{strict} ^{lo}	FDP _{strict} ^{hi}
e-LOND/uniform	151	0.026	0.007	0.000	0.026
ADDIS/poly	152	0.033	0.007	0.000	0.026
union	152	0.033	0.007	0.000	0.026

TABLE XXXVIII

Bates calibration-conditional adjustment ($n_{\text{cal}} = 1000$): it fixes conditional validity at a power cost set by the same quantity. The first three columns are the conditional exceedance probability — analytic prediction, measured nominal, and β -adjusted; the last two are the e-LOND detection cost (median rejections, nominal vs β -adjusted). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order table I reports (table X); the order shifts detections and the level an alert fires at, and shifts nothing else. The β -bound requires 0.22–0.62 of $|C|$ across error targets $\delta \in [0.01, 0.2]$.

k	conditional exceedance			e-LOND rej. (med)	
	predicted	nominal	β -adj.	nominal	β -adj.
1	0.368	0.370	0.092	100	62.5
10	0.459	0.439	0.084	0	0
100	0.489	0.483	0.092	0	0